



US012410578B2

(12) **United States Patent**
Horii et al.

(10) **Patent No.:** **US 12,410,578 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **WORKING MACHINE**

(71) Applicant: **KUBOTA CORPORATION**, Osaka
(JP)

(72) Inventors: **Hiroshi Horii**, Sakai (JP); **Hiroya Mori**, Sakai (JP); **Atsushi Matsumoto**, Sakai (JP)

(73) Assignee: **KUBOTA CORPORATION**, Osaka
(JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 759 days.

(21) Appl. No.: **17/325,826**

(22) Filed: **May 20, 2021**

(65) **Prior Publication Data**

US 2021/0270003 A1 Sep. 2, 2021

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2019/050491, filed on Dec. 24, 2019.

(30) **Foreign Application Priority Data**

Dec. 25, 2018 (JP) 2018-241470
Jun. 27, 2019 (JP) 2019-120305

(51) **Int. Cl.**
E02F 3/32 (2006.01)
E02F 9/20 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **E02F 3/325** (2013.01); **E02F 9/2271**
(2013.01); **E02F 9/264** (2013.01); **F15B 15/16**
(2013.01);
(Continued)

(58) **Field of Classification Search**
CPC . E02F 3/325; E02F 3/964; E02F 9/006; E02F 9/2004; E02F 9/2203; E02F 9/2271;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2002/0062587 A1 5/2002 Kimoto et al.
2013/0345939 A1* 12/2013 Magaki E02F 9/2075
701/50

(Continued)

FOREIGN PATENT DOCUMENTS

CN 101243259 A 8/2008
CN 104583178 A 4/2015

(Continued)

OTHER PUBLICATIONS

Extended European Search Report for European Application No. 19906478.3, dated Sep. 28, 2022.

Primary Examiner — Lisa M Caputo

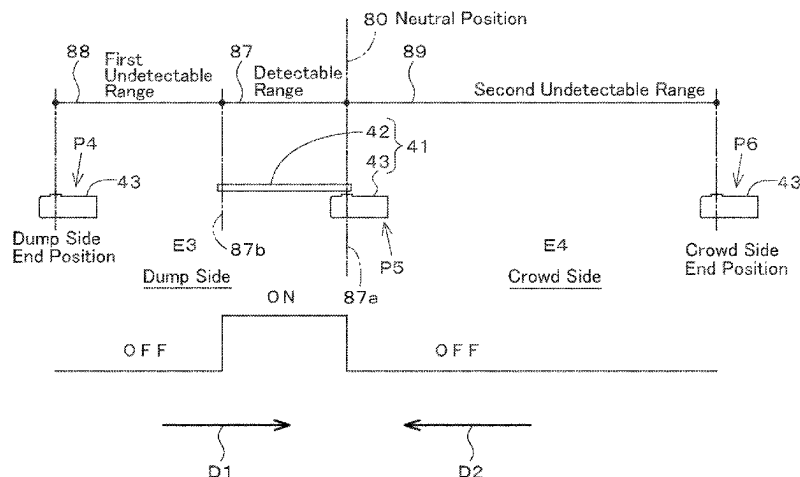
Assistant Examiner — Joshua L Forristall

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch & Birch, LLP

(57) **ABSTRACT**

A working machine, in which a detection device for detecting a swing position of a supported member is prevented from being damaged, includes a support member pivotally supporting the supported member via a pivot, and a cylinder telescopically moved by extension and contraction movement of its piston rod with respect to its cylinder tube by hydraulic fluid flowing through a fluid passage formed in the piston rod. The cylinder tube is pivotally supported by a first end portion of the support member, and the piston rod is pivotally supported by a second end portion of the support member. The detection device for detecting a telescopic movement state of the cylinder is provided between the cylinder and the support member.

16 Claims, 23 Drawing Sheets



US 12,410,578 B2

Page 2

(51)	Int. Cl.		2018/0178342 A1 *	6/2018	Russell		G01C 9/04
	E02F 9/22	(2006.01)	2019/0085875 A1 *	3/2019	Slama		G01D 5/12
	E02F 9/26	(2006.01)	2020/0040554 A1 *	2/2020	Nishimura		E02F 9/264
	F15B 15/16	(2006.01)	2020/0173148 A1 *	6/2020	Nishi		E02F 9/26
(52)	U.S. Cl.		2021/0002853 A1 *	1/2021	Yamashita		E02F 9/2203
	CPC	 E02F 9/2004 (2013.01); E02F 9/2203 (2013.01); E02F 9/2278 (2013.01)	2021/0148082 A1 *	5/2021	Suzuki		E02F 9/265
			2021/0230827 A1 *	7/2021	Bitter		E02F 3/435
(58)	Field of Classification Search		FOREIGN PATENT DOCUMENTS				
	CPC	 E02F 9/2278; E02F 9/264; F15B 15/16; F15B 15/2807; F15B 15/2861; F15B 15/2892	DE	20	2013 003 314 U1	8/2014	
	USPC	 702/150	EP		1942281 A1	7/2008	
	See application file for complete search history.		JP		10-121523 A	5/1998	
			JP		2006-220179 A	8/2006	
			JP		2010-168775 A	8/2010	
			JP		2011-252338 A	12/2011	
			JP		2013133913 A *	7/2013	 F15B 15/149
			JP		2013-167099 A	8/2013	
			JP		2018-146726 A	9/2018	
			WO	WO	2015/060730 A1	4/2015	
(56)	References Cited						
	U.S. PATENT DOCUMENTS						
	2016/0138240 A1 *	5/2016 Ikegami		E02F 9/26			
				60/420			
	2017/0260717 A1 *	9/2017 Seki		E02F 3/32			
					* cited by examiner		

Fig.1

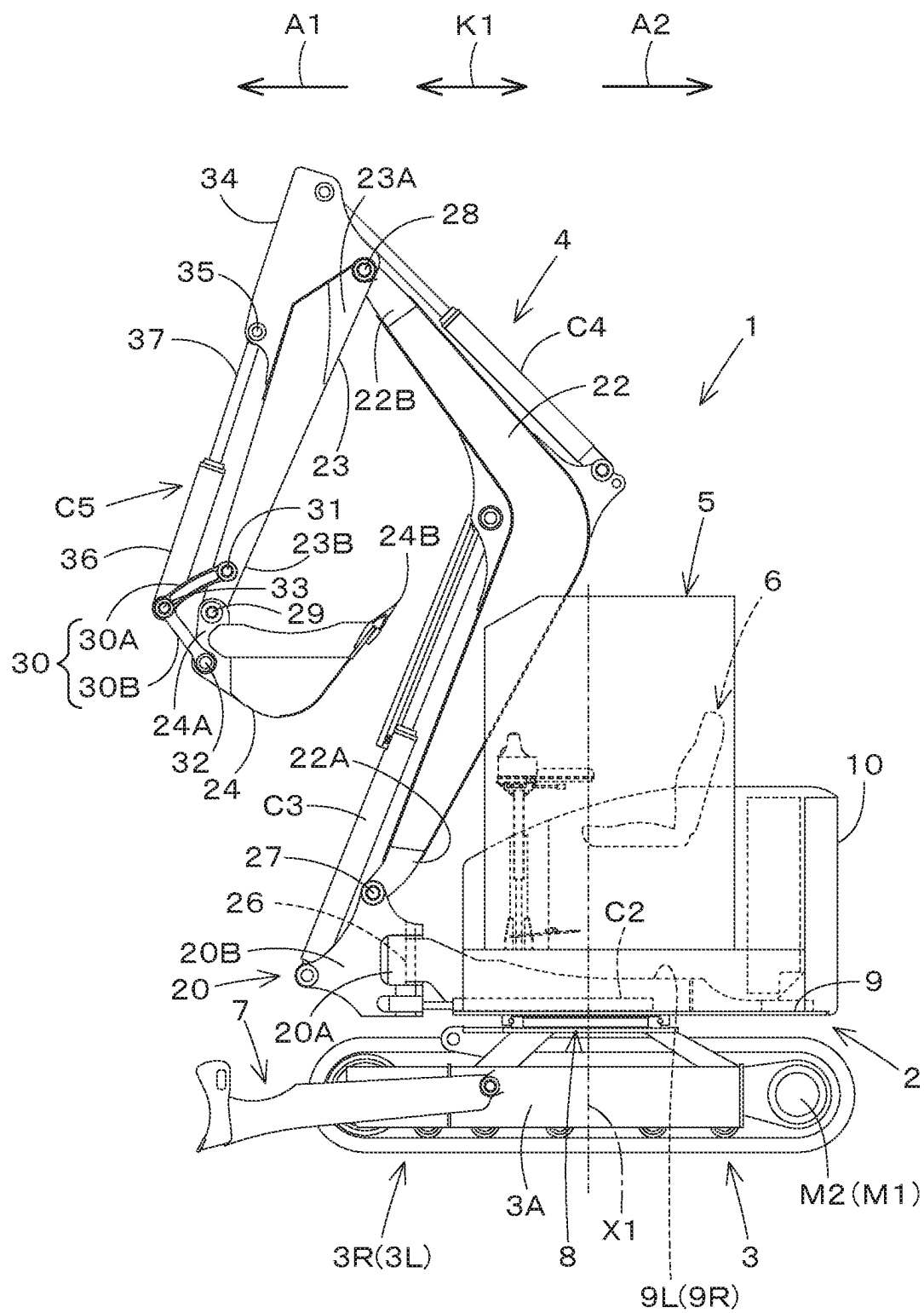


Fig.2

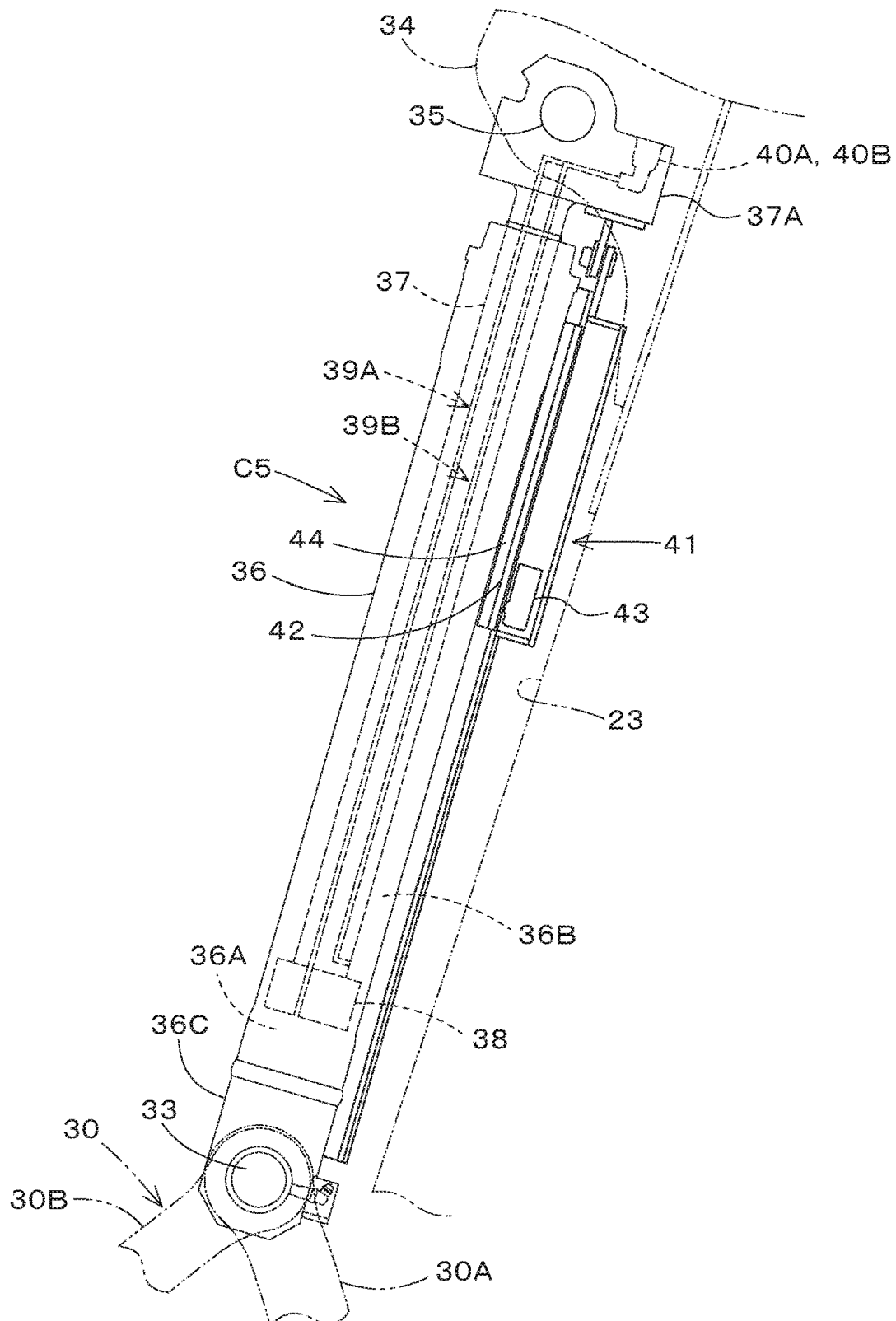


Fig.3

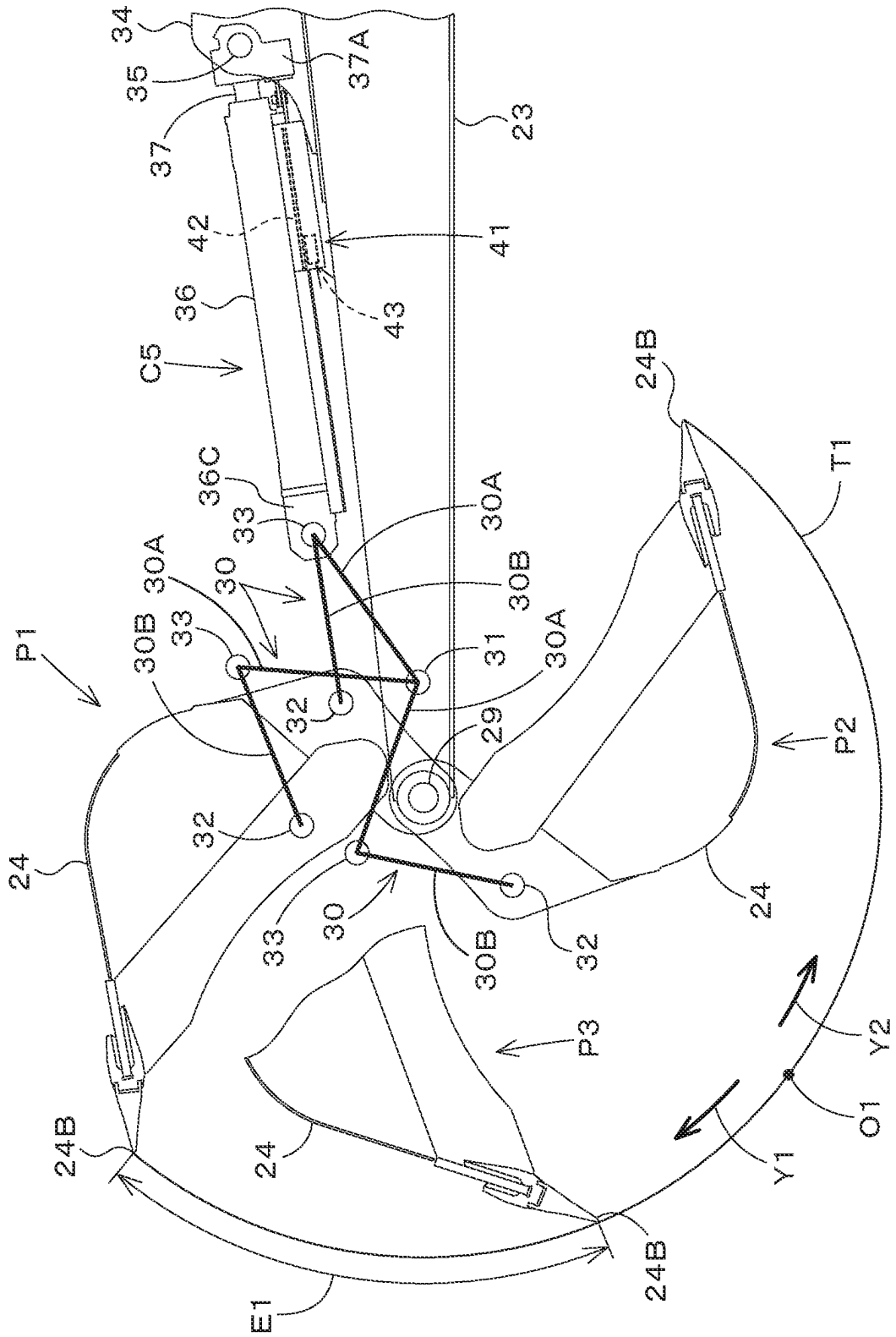


Fig.4

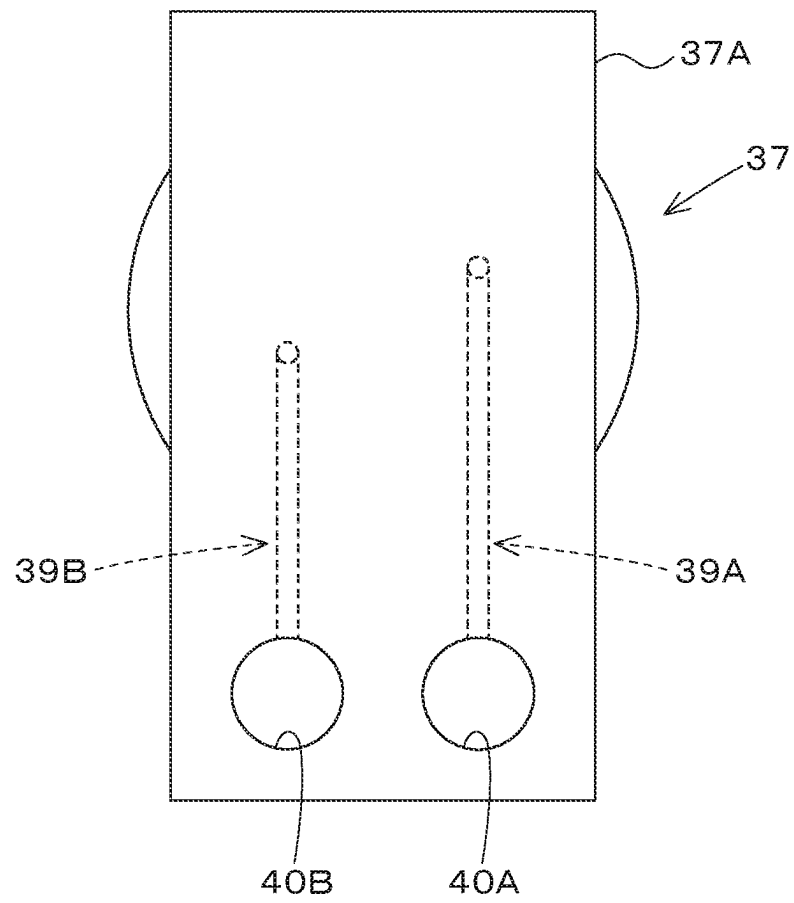


Fig. 5

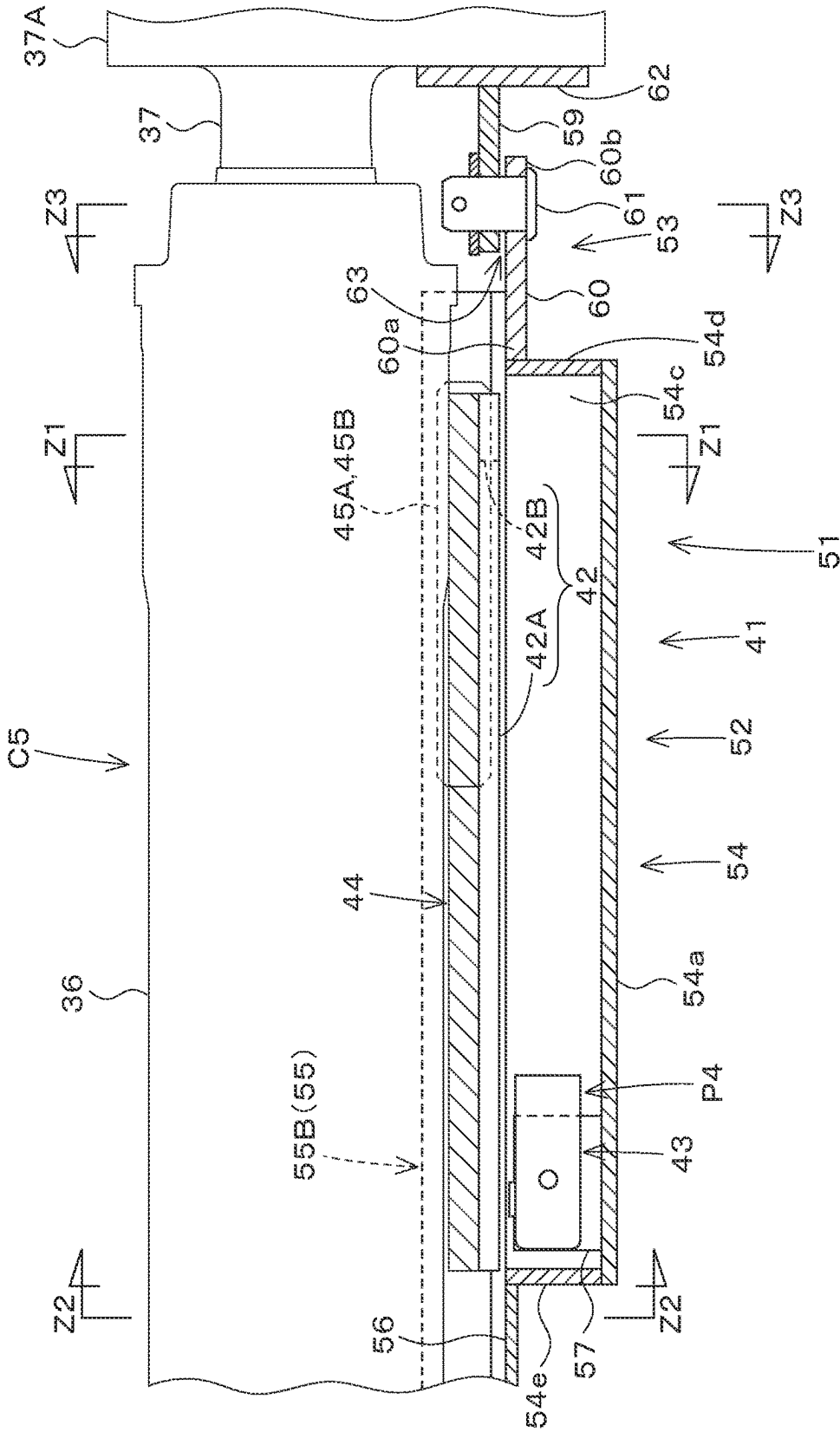


Fig. 6

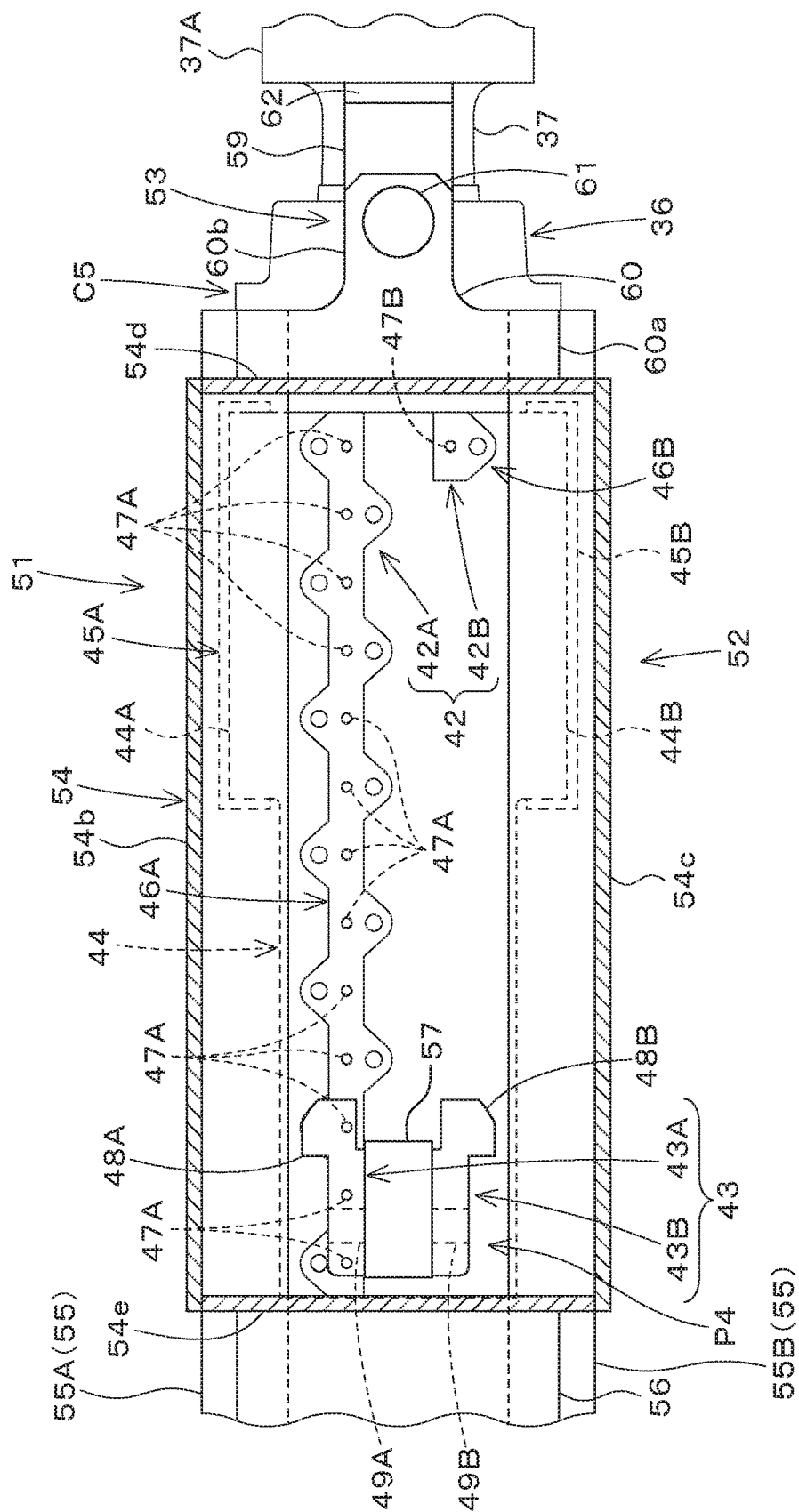


Fig. 7

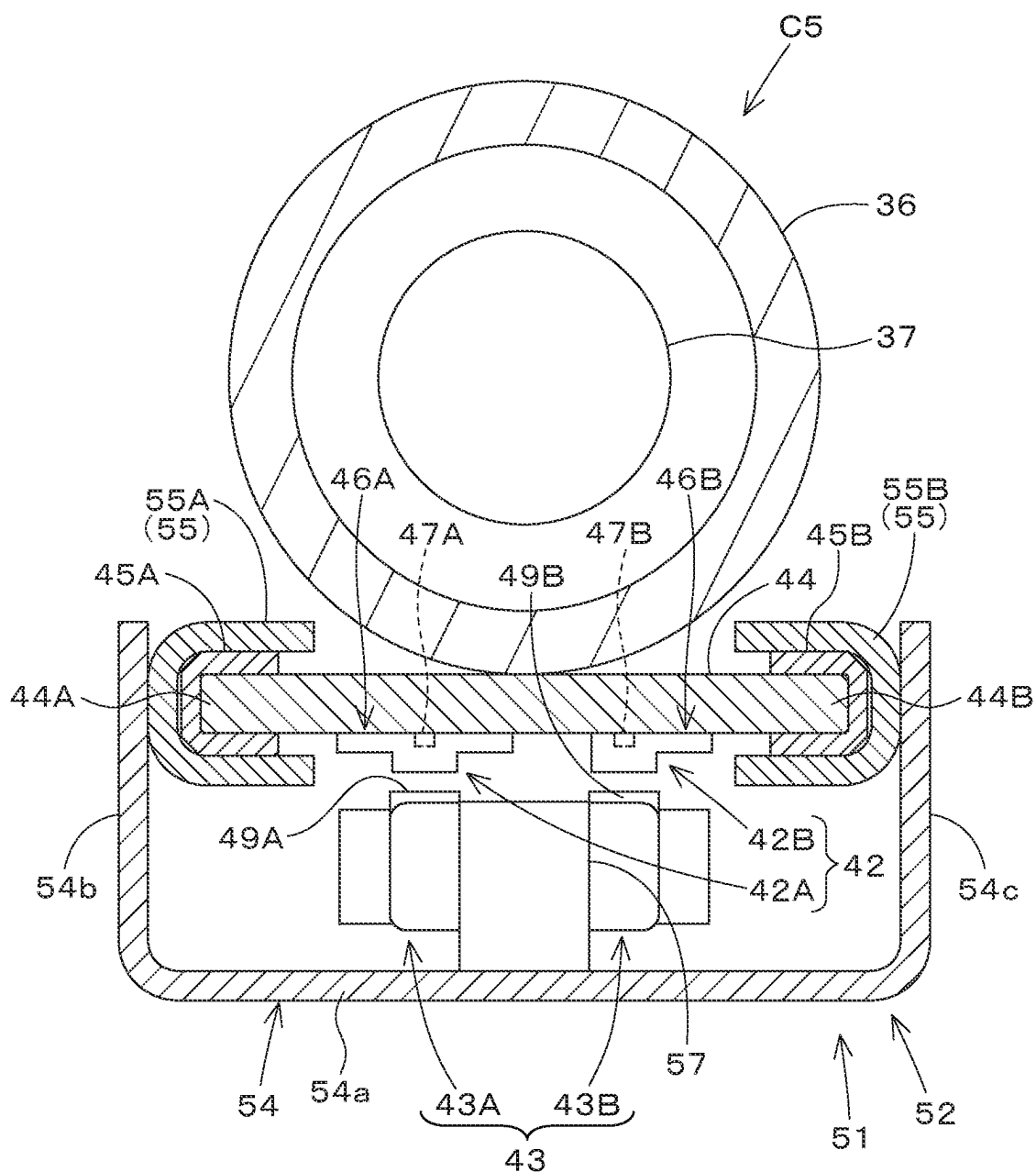


Fig.8

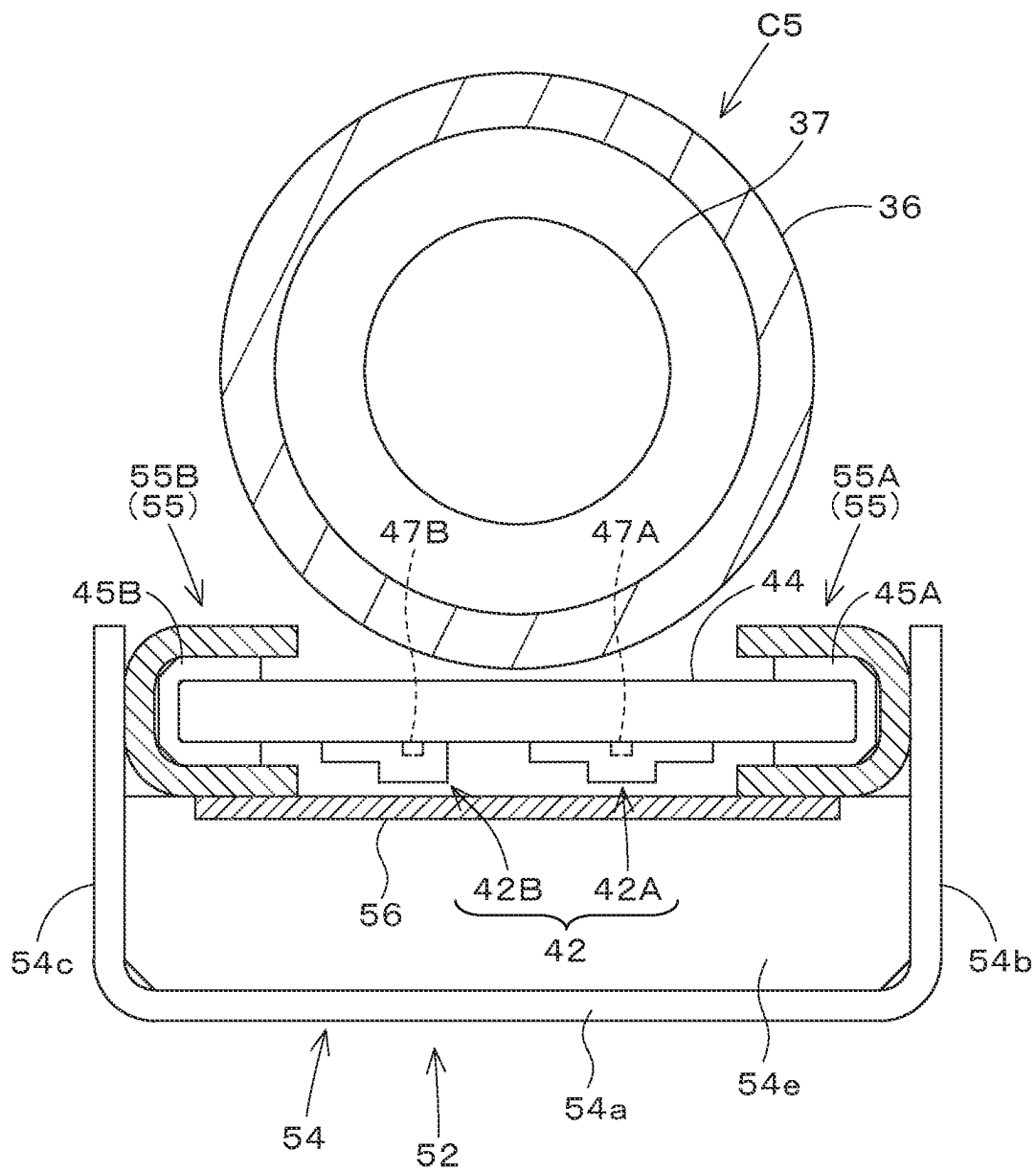


Fig. 9

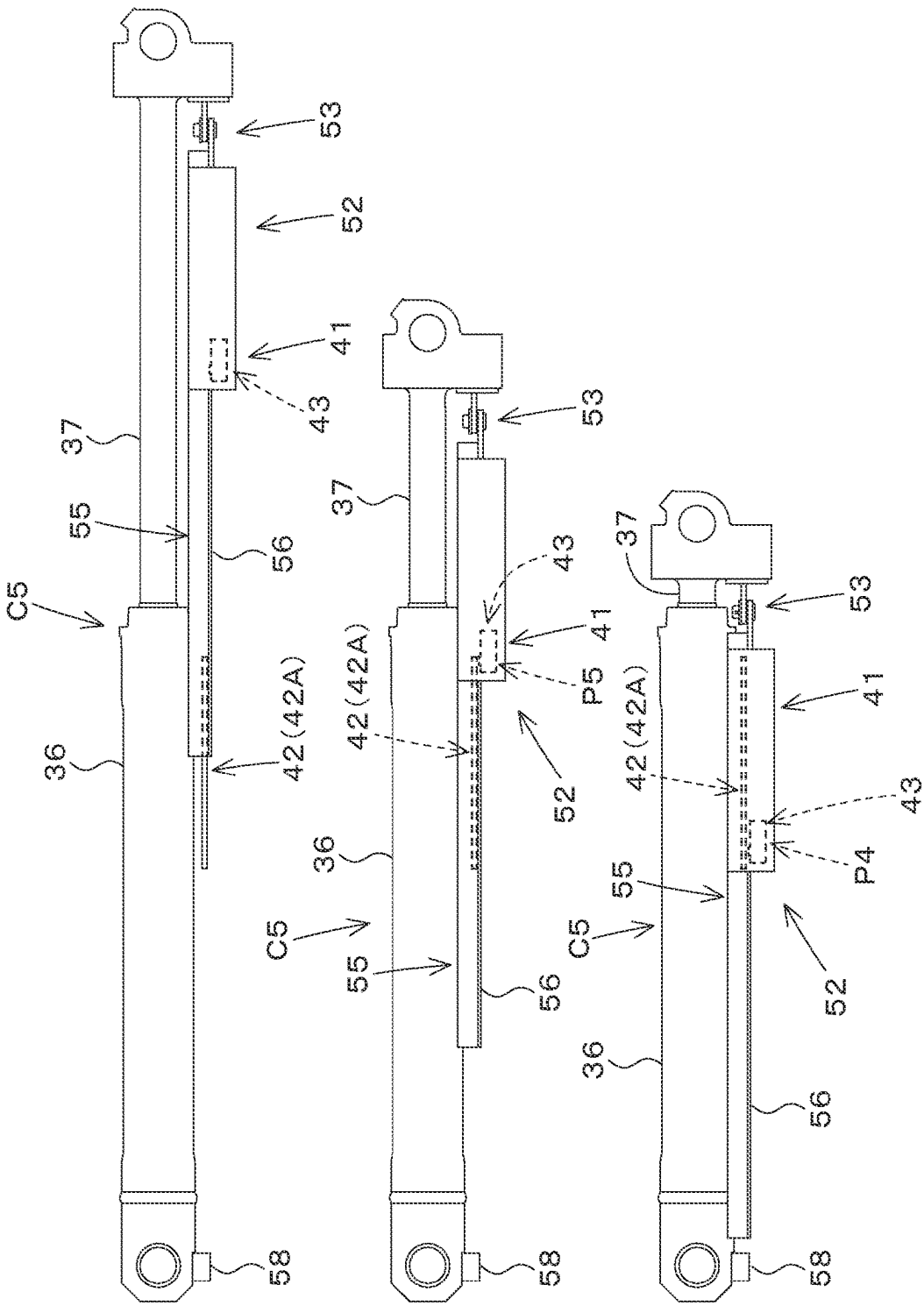
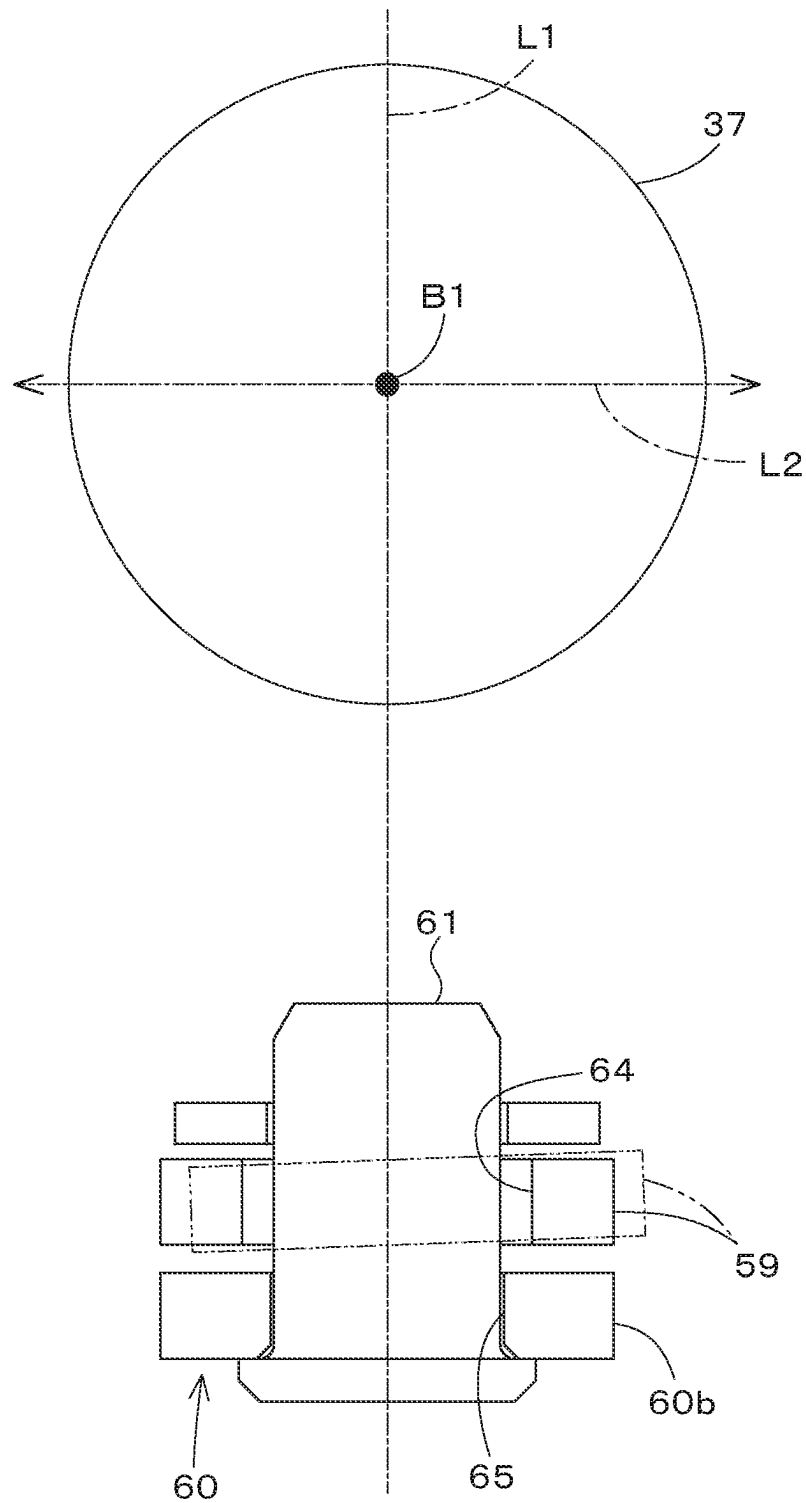
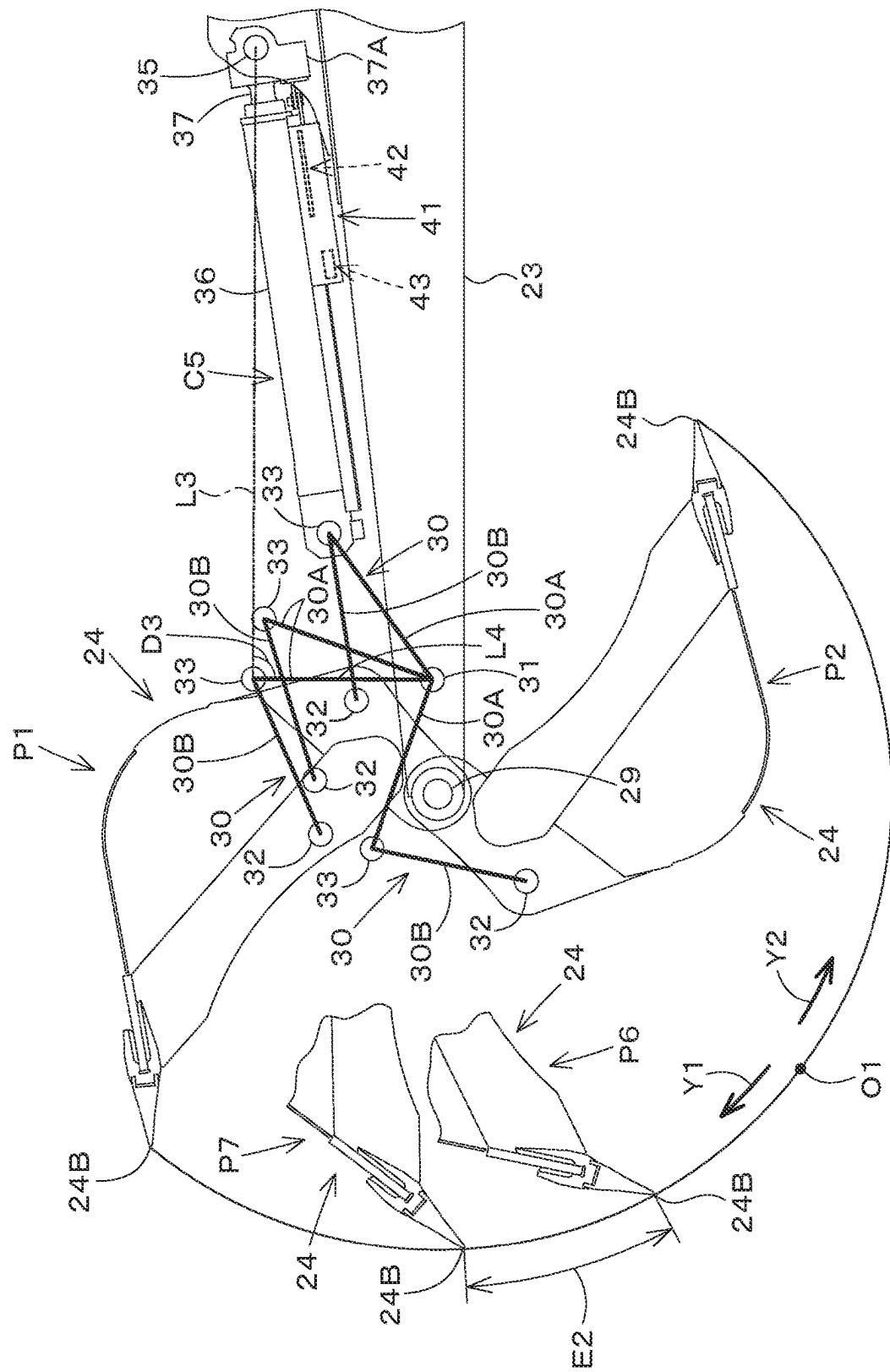


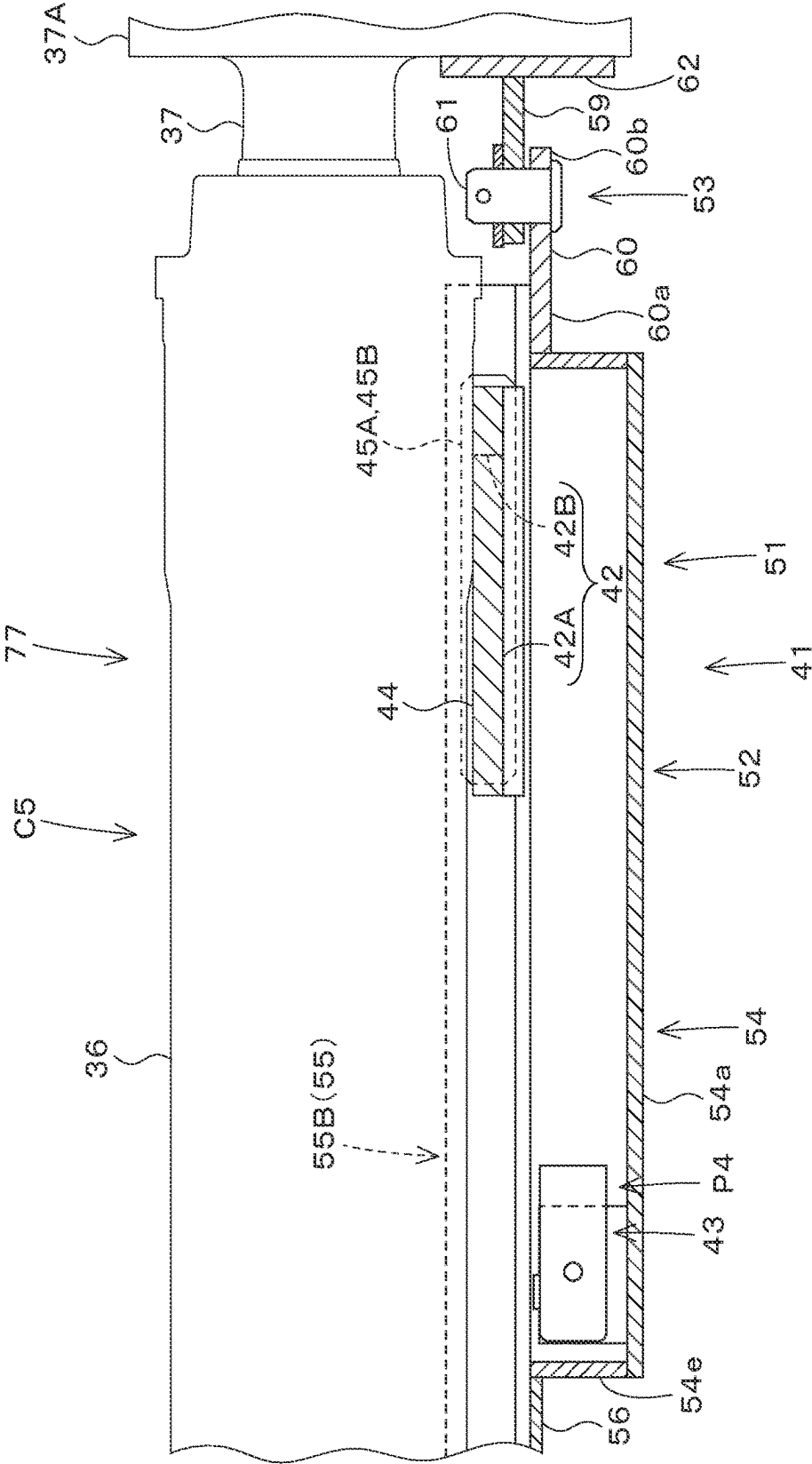
Fig.10





Y
Y
o
x
L

Fig.12




 Կ
 Ե
 Լ
 Լ

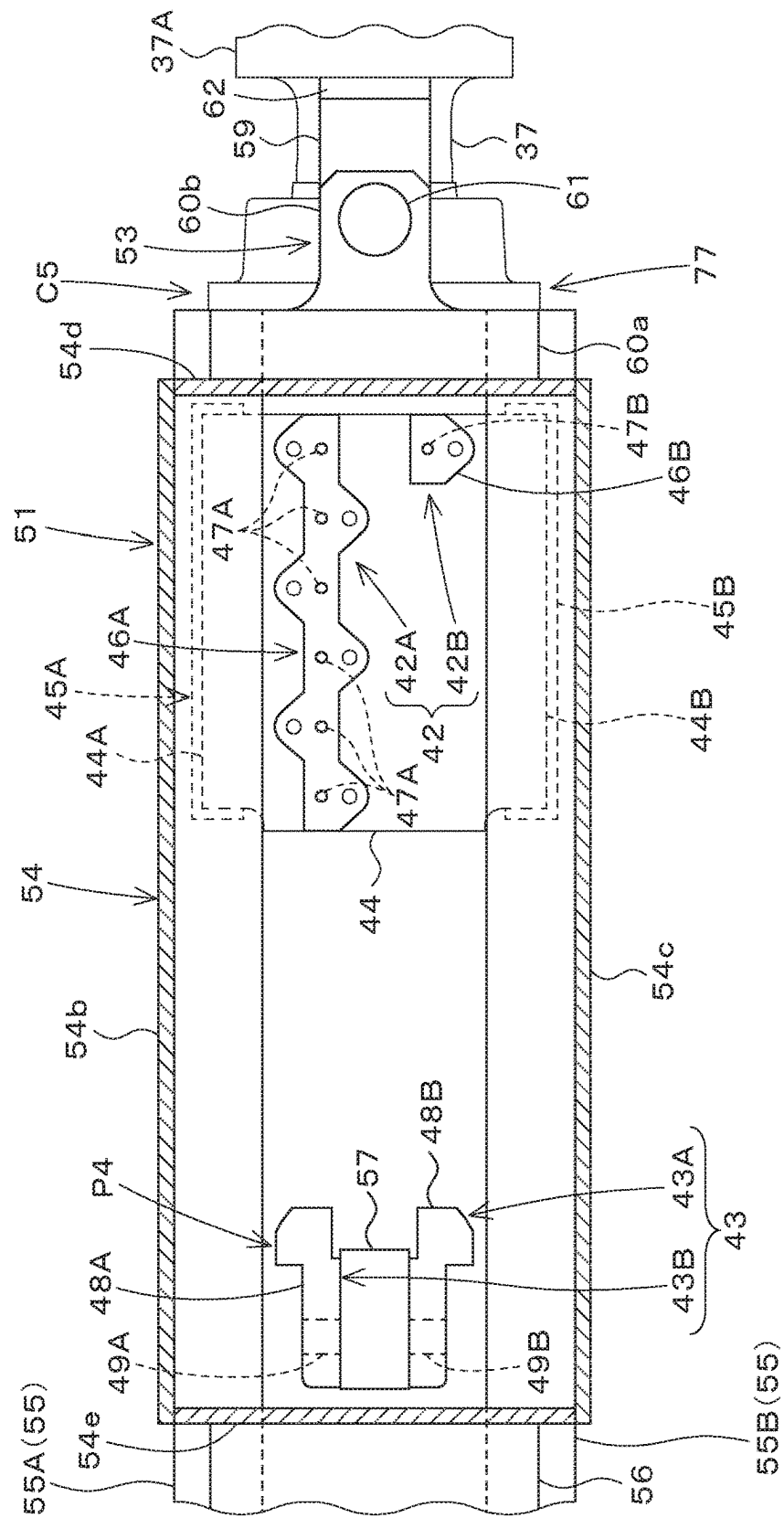


Fig. 14

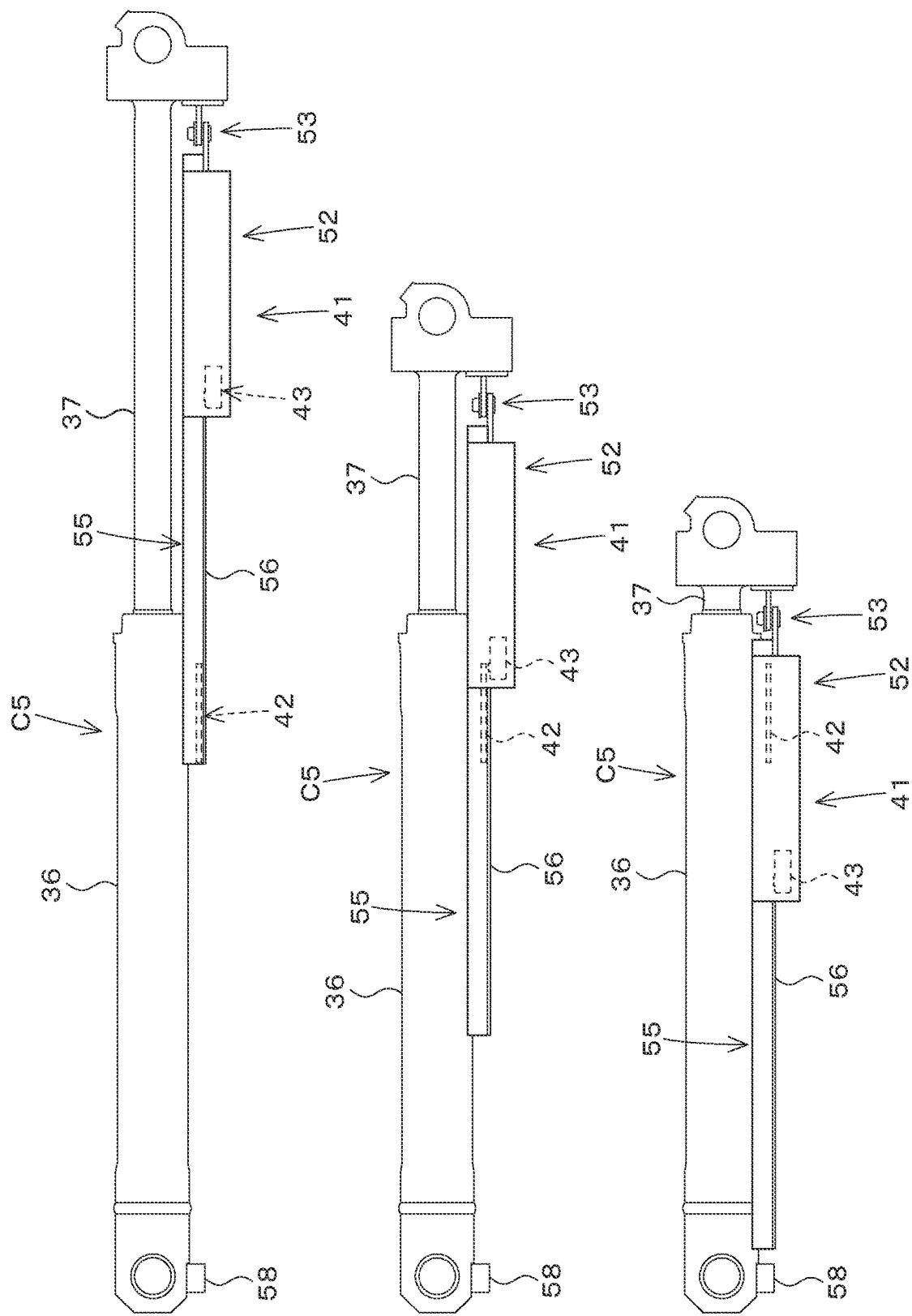


Fig.15

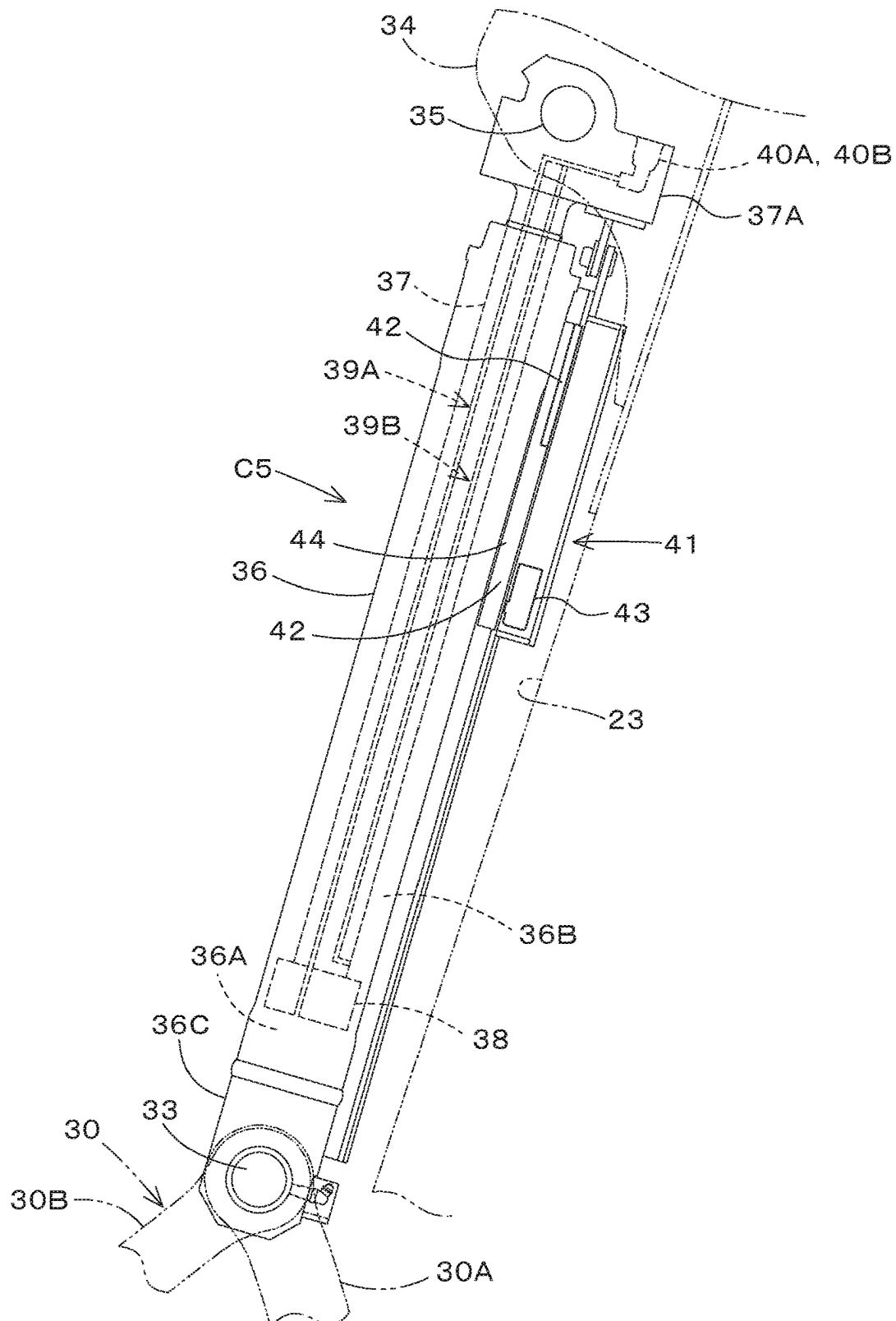


Fig. 16

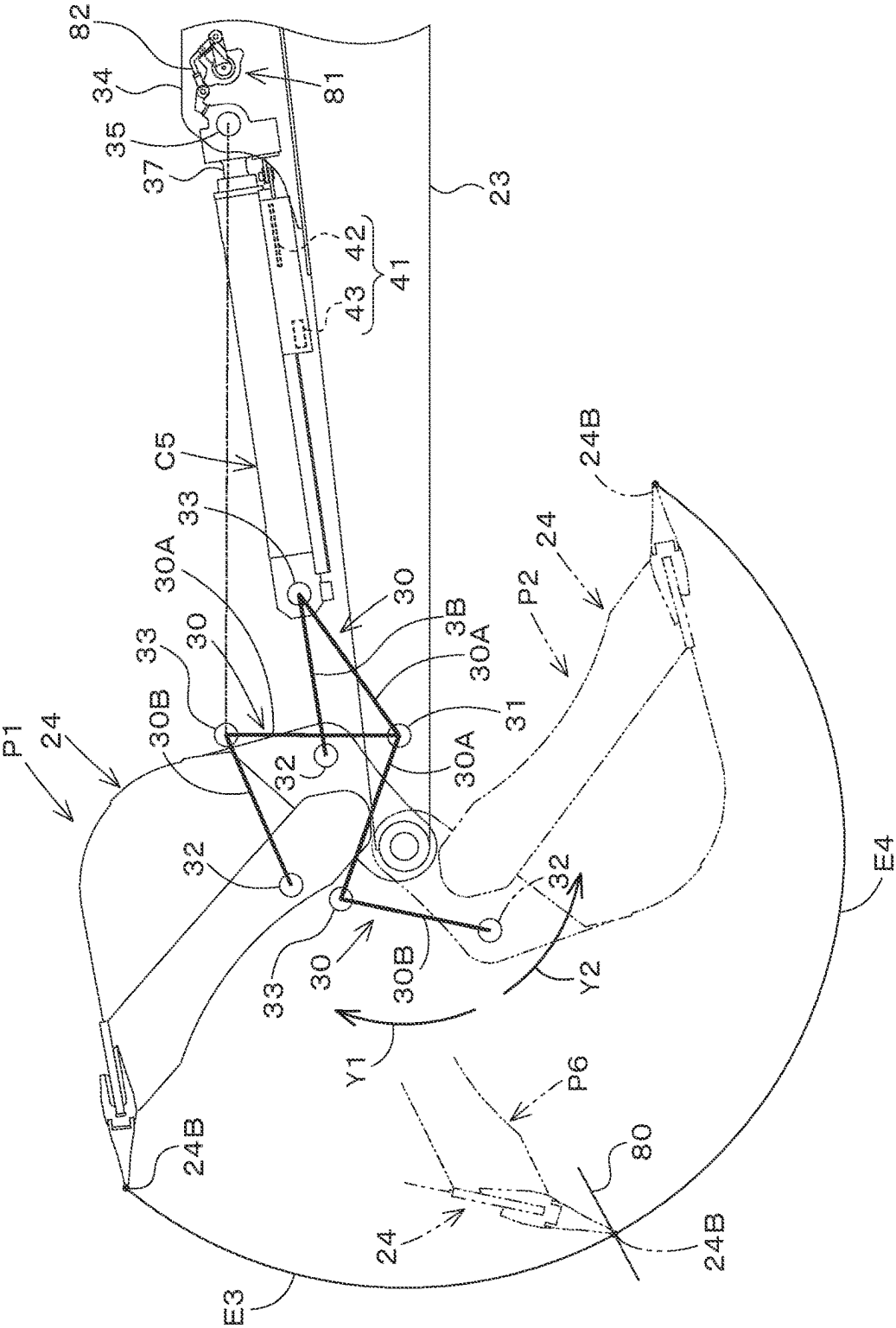
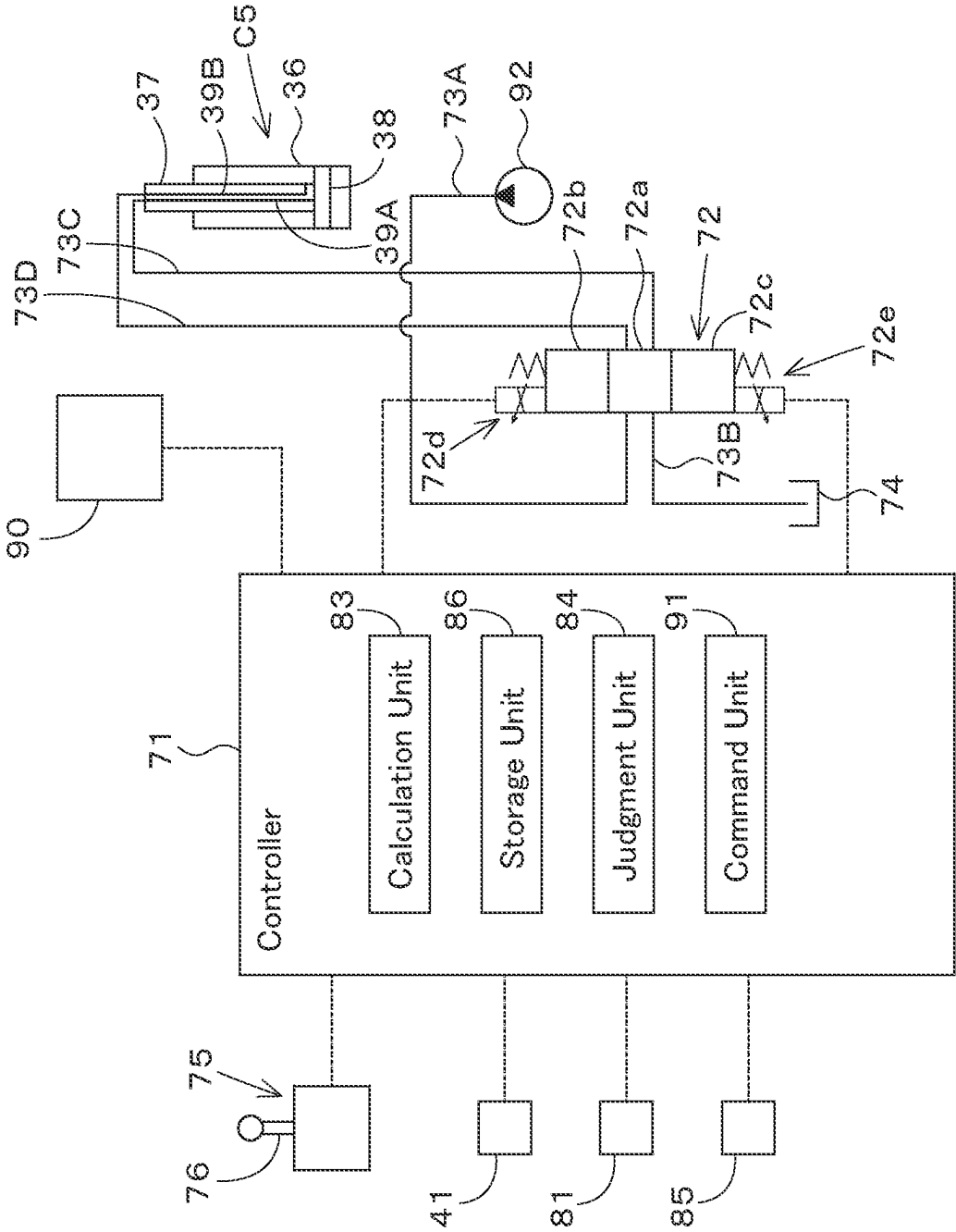


Fig.17





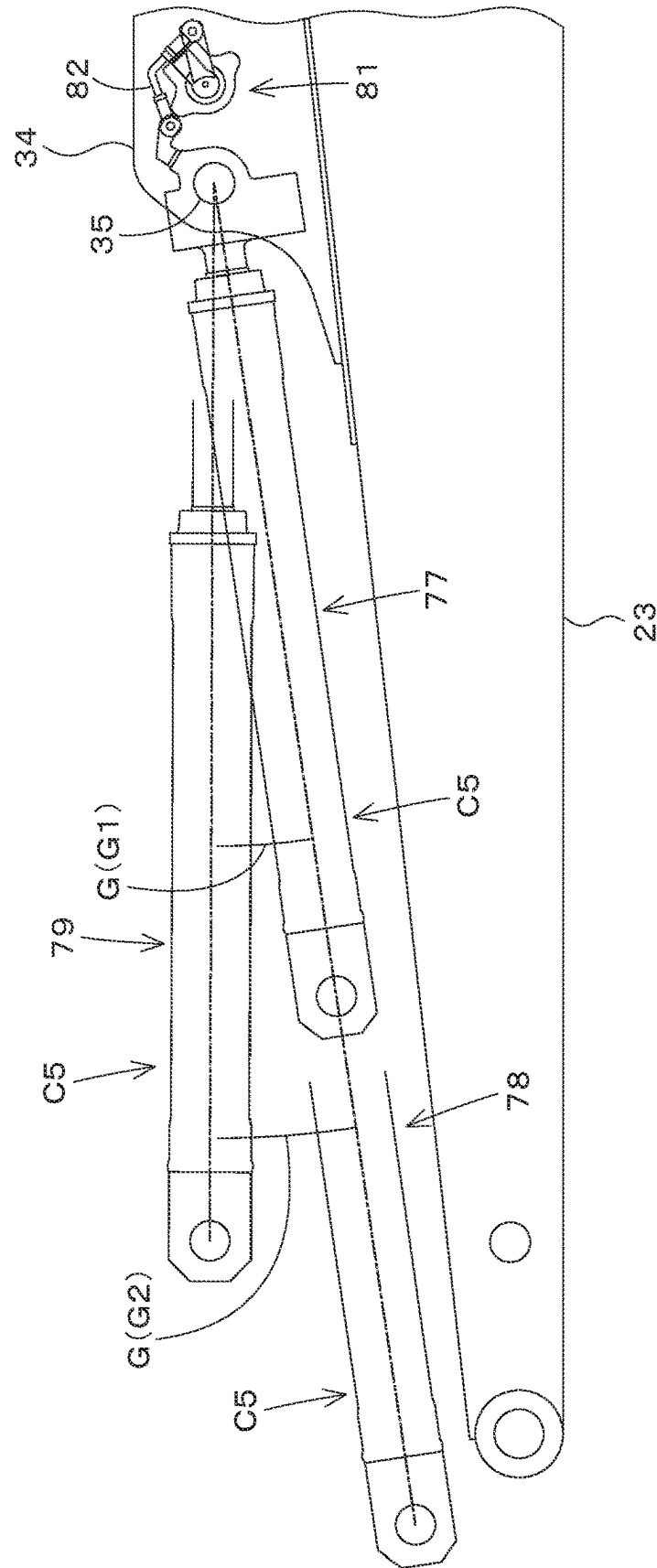


Fig. 19

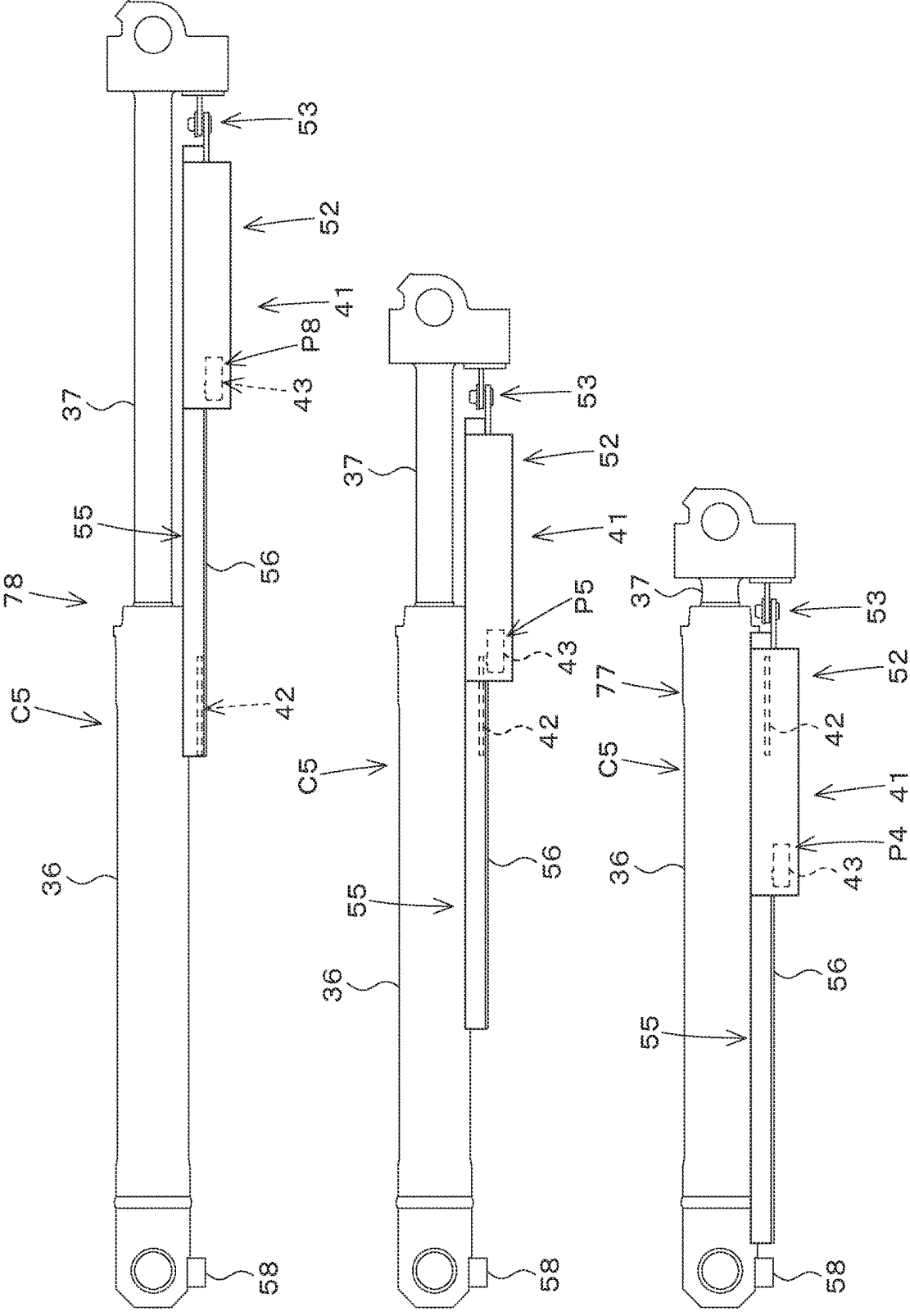


Fig.20

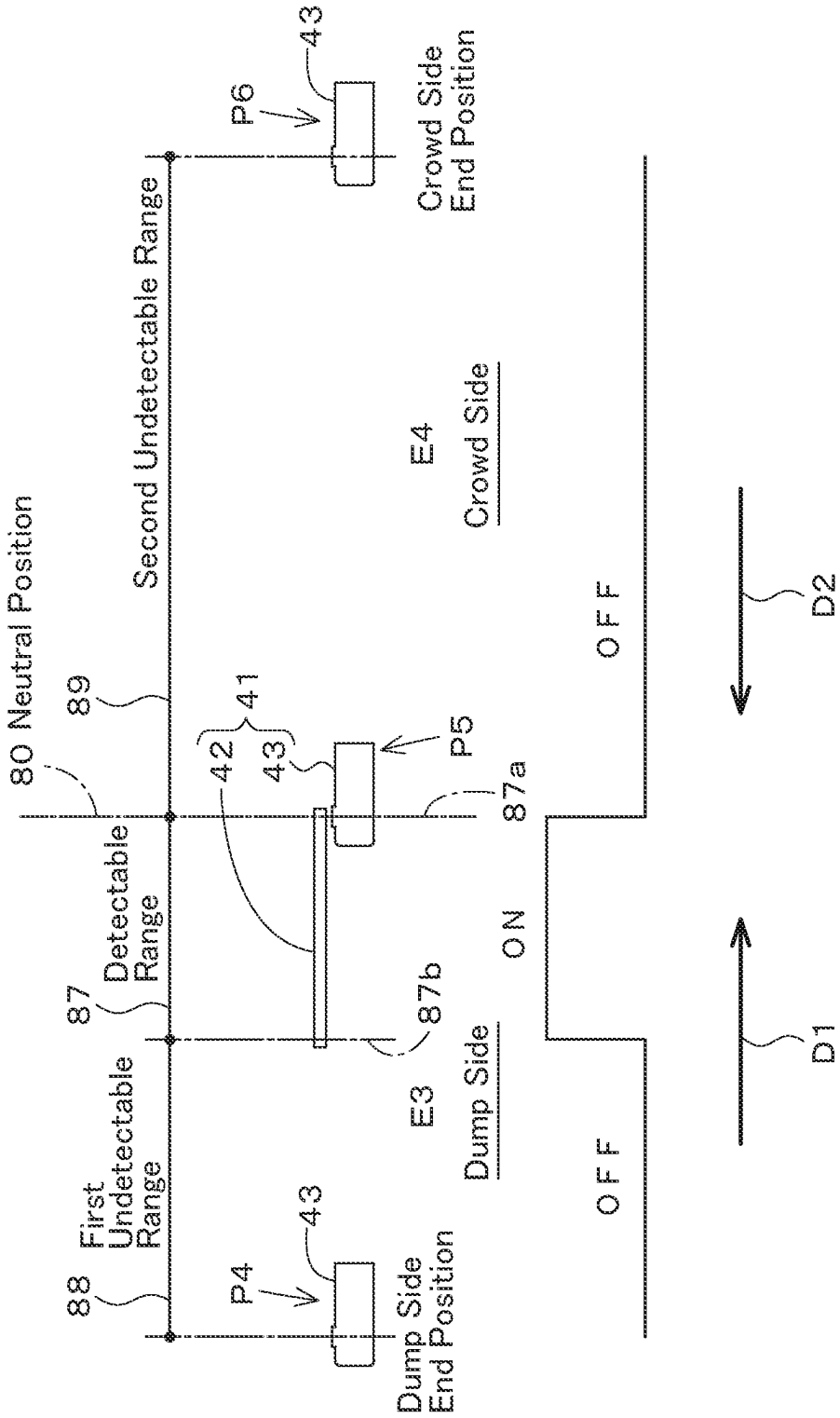


Fig.21

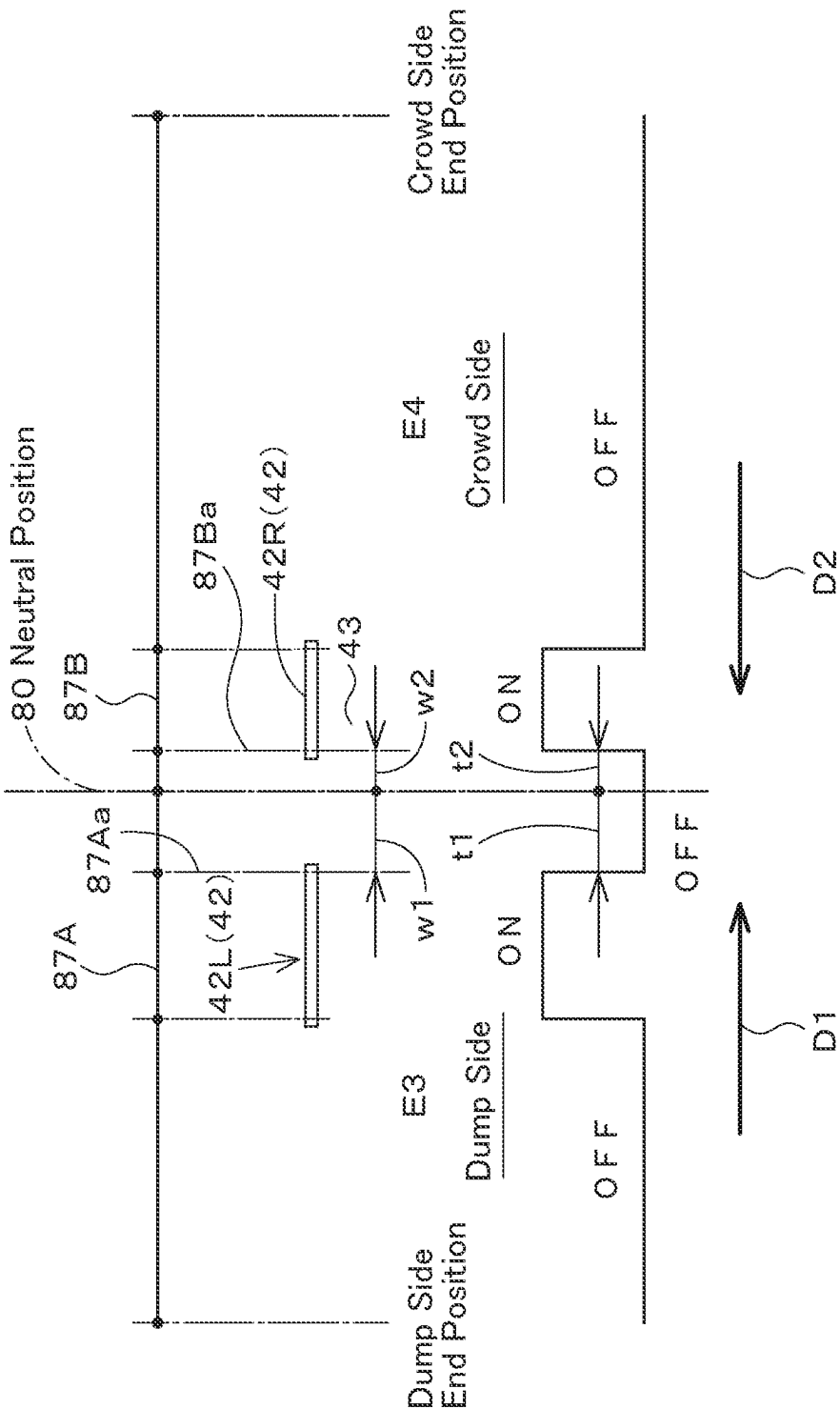


Fig.22

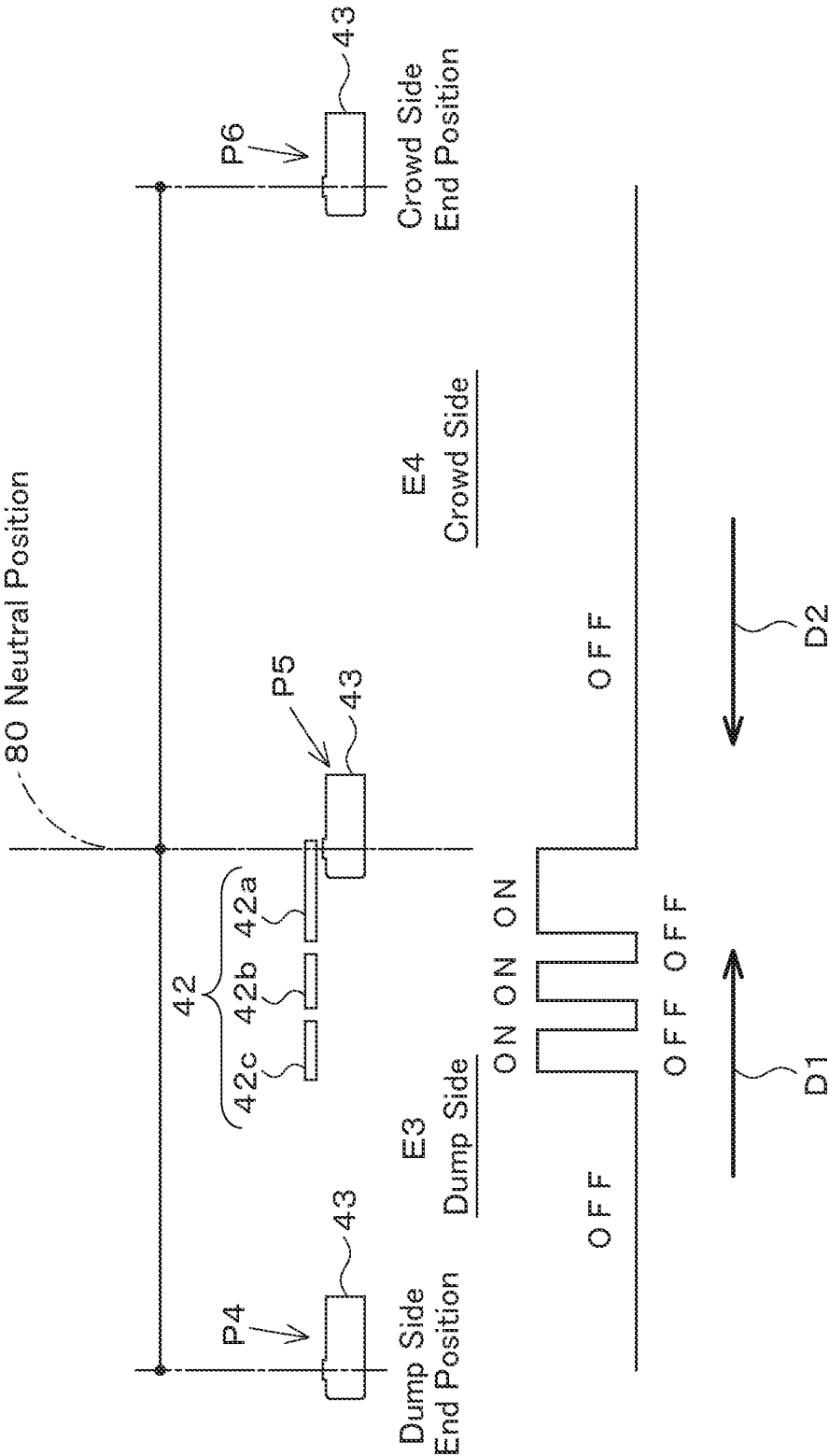
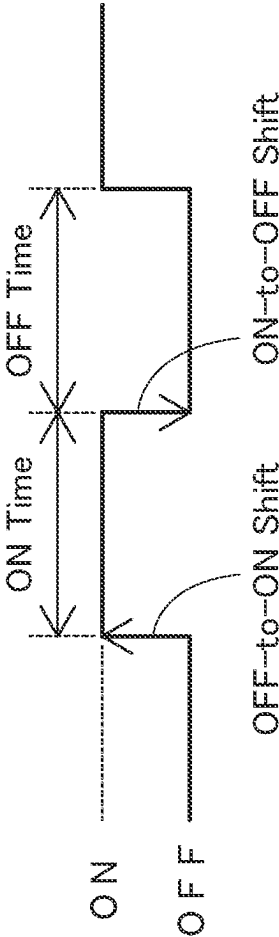


Fig.23

	First Direction	Second Direction
OFF→ON	OFF-to-ON Shifts Count B1	OFF-to-ON Shifts Count F1
ON→OFF	ON-to-OFF Shifts Count B2	ON-to-OFF Shifts Count F2
ON Time	Time B3	Time F3
OFF Time	Time B4	Time F4



1

WORKING MACHINE**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation application of International Application No. PCT/JP2019/050491, filed on Dec. 24, 2019, which claims the benefit of priority to Japanese Patent Application No. 2018/241470, filed on Dec. 25, 2018 and to Japanese Patent Application No. 2019/120305, filed on Jun. 27, 2019. The entire contents of each of these applications are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION**Field of the Invention**

The present invention relates to a working machine such as a backhoe.

Description of the Related Art

The working machine disclosed in Japanese Unexamined Patent Publication No. 2011-252338 is known.

The working machine disclosed in Japanese Unexamined Patent Publication No. 2011-252338 includes a working device attached to a machine body thereof. The working device includes a boom swingably attached to the machine body, an arm swingably supported at a basal end portion thereof by the boom, and a working tool (e.g., a bucket) pivotally supported by a tip end portion of the arm via a pivot. A bucket cylinder (serving as a working tool cylinder) is pivotally supported by the arm via a cylinder shaft. The bucket is swung to a dump side and to a crowd side by telescoping the bucket cylinder.

The working machine includes a detection device for detecting the swing position of the working tool around the pivot. The detection device is provided on a pivotally supported portion of the working tool via the pivot.

SUMMARY OF THE INVENTION

In a first aspect of the present invention, a working machine comprises a support member, a supported member pivotally supported by an end portion of the support member via a pivot, and a cylinder of which telescopic movement causes the supported member to swing around the pivot. The cylinder includes a cylinder tube, and a piston rod inserted into the cylinder tube. The cylinder is constituted by a hydraulic cylinder telescopically moved by extension and contraction movement of the piston rod with respect to the cylinder tube by hydraulic fluid flowing through a fluid passage formed in the piston rod. The cylinder tube is pivotally supported by one of end portions of the support member, and the piston rod is pivotally supported by the other of the end portions of the support member. A detection device for detecting a telescopic movement state of the cylinder is provided between the cylinder and the support member.

The working machine comprises a machine body, and a boom swingably attached to the machine body. The support member is an arm pivotally supported at a basal end portion thereof by the boom. The supported member is a working tool pivotally supported by a tip end portion of the arm via the pivot. The cylinder is a working tool cylinder for swinging the working tool around the pivot.

2

The detection device includes a detected object provided on one of the cylinder tube and the piston rod, and a detector provided on the other of the cylinder tube and the piston rod. The detector detects the detected object for detection of the telescopic movement state of the cylinder.

The detected object is attached onto an outer surface of the cylinder tube facing the support member. The detected object has a predetermined length in the longitudinal direction of the cylinder tube, and includes a magnet. The detector is constituted by a proximity sensor, which is attached to the piston rod between the cylinder tube and the support member so as to be able to face the detected object, and which is configured to move together with the piston rod to detect the magnet.

The working machine comprises a detector mount to which the detector is attached; and a connection mechanism connecting the detector mount to the piston rod. The piston rod includes a first connection piece joined to the piston rod, a second connection piece joined to the detection attachment member, and a connection pin passed through the first connection piece and the second connection piece so as to connect the first connection piece and the second connection piece to each other. A pin hole is formed in the connection piece to allow the connection pin to pass therethrough. The pin hole is formed into a long hole shape to allow axial rotation of the piston rod.

The working machine comprises an arm serving as the support member, a working tool serving as the supported member pivotally and swingably supported by a tip end portion of the arm, a working tool cylinder serving as the cylinder pivotally supported by the arm via a cylinder shaft, the telescopic movement of the working tool cylinder causing the working tool to swing, an angle sensor which detects a swing angle of the working tool cylinder when the working tool is disposed on a dump side with respect to a neutral position where a swing angle of the working tool cylinder around the cylinder shaft becomes maximum, and which detects a swing angle of the working tool cylinder when the working tool is disposed on a crowd side with respect to the neutral position, and a controller for determining the swing position of the working tool based on the swing angle detected by the means of angle sensor and a result of detection by means of the detection device. The detection device detects ON and OFF signals indicating whether the working tool cylinder is extended further from the neutral position or is contracted further from the neutral position. The controller determines whether the working tool is on the dump side or the crowd side in accordance with a first detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a first direction from the dump side to the crowd side, and a second detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a second direction from the crowd side to the dump side.

In a second aspect of the invention, a working machine comprises an arm, a working tool pivotally and swingably supported by a tip end portion of the arm; a working tool cylinder pivotally supported by the arm via a cylinder shaft, the telescopic movement of the working tool cylinder causing the working tool to swing, an angle sensor which detects a swing angle of the working tool cylinder when the working tool is disposed on a dump side with respect to a neutral position where a swing angle of the working tool cylinder around the cylinder shaft becomes maximum, and which detects a swing angle of the working tool when the working tool is disposed on a crowd side with respect to the neutral

3

position, a detection device which detects ON and OFF signals indicating whether the working tool cylinder is extended further from the neutral position or is contracted further from the neutral position, and a controller for determining the swing position of the working tool based on the swing angle detected by means of the angle sensor and a result of detection by means of the detection device. The controller determines whether the working tool is on the dump side or the crowd side in accordance with a first detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a first direction from the dump side to the crowd side, and a second detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a second direction from the crowd side to the dump side.

Based on the first detection pattern and the second detection pattern, the controller determines whether the working tool, when in a predetermined range as the vicinity of the neutral position, is on the dump side or the crowd side.

While the working tool is disposed out of the predetermined range, the controller keeps a result of the determination regarding the position of the working tool when in the predetermined range.

The working tool cylinder includes a cylinder tube, and a piston rod extended and contracted with respect to the cylinder tube. The detection device includes a detected object provided on one of the cylinder tube and the piston rod, and a detector provided on the other of the cylinder tube and the piston rod. The detector outputs the on or off signal in response to whether the detected object is detected or not.

Either a range ranging from the neutral position to an intermediate position between the neutral position and an end position on the dump side or a range ranging from the neutral position to an intermediate position between the neutral position and an end position on the crowd side is defined as a detectable range where the detected object is detectable by means of the detector.

Each of the first detection pattern and the second detection pattern is a combination of ON-to-OFF and OFF-to-ON signal shifts. The combination as the first detection pattern and the combination as the second detection pattern are different from each other.

When operation of the working tool is ended, the controller stores the result of determination of whether the working tool is on the dump side or the crowd side. The controller defines the position of the working tool stored on the ending of operation of the working tool as an initial position for restarting of operation of the working tool, and performs the determination when the working tool cylinder is extended or contracted from the initial position.

The working machine comprises a display unit connected to the controller. The display unit displays a message for urging an operator to operate to locate the working tool on either the dump side or the crowd side.

The controller performs the determination based on a detection value outputted from the angle sensor when an ON/OFF signal shift occurs in detection by means of the detection device.

When operation of the working tool is ended, the controller stores information indicating whether the working tool is on the dump side or the crowd side. When operation of the working tool is restarted, the controller determines the swing position of the working tool based on the information stored on the last ending of operation of the working tool.

The working machine comprises an instruction input unit for receiving an instruction input from an operator, the

4

instruction input indicating whether the working tool is on the dump side or the crowd side.

The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete appreciation of preferred embodiments of the present invention and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings described below.

FIG. 1 is a side view of a working machine.

FIG. 2 is a side view of a working tool cylinder.

FIG. 3 is a side view of a working tool in operation.

FIG. 4 is a view of a rod head of a piston rod when viewed axially.

FIG. 5 is a sectional side view of a detection device.

FIG. 6 is a view of the detection device when viewed from an arm side.

FIG. 7 is a cross sectional view taken along Z1-Z1 line of FIG. 5.

FIG. 8 is a cross sectional view taken along Z2-Z2 line of FIG. 5.

FIG. 9 is a side view of the working tool cylinder in telescoping movement.

FIG. 10 is a cross sectional view taken along Z3-Z3 line of FIG. 5.

FIG. 11 is a side view of a working tool in operation according to an alternative embodiment.

FIG. 12 is a sectional side view of a detection device arranged according to the alternative embodiment.

FIG. 13 is a sectional side view of the detection device arranged according to the alternative embodiment.

FIG. 14 illustrates side views of a working cylinder according to the alternative embodiment in different telescopic movement states.

FIG. 15 is a side view of the working tool cylinder.

FIG. 16 is a side view of the working tool in operation.

FIG. 17 is a schematic diagram of a control system.

FIG. 18 illustrates side views of the working cylinder in different swing states.

FIG. 19 illustrates side views of the working cylinder in different telescopic movement states.

FIG. 20 is a diagram describing detection patterns.

FIG. 21 is a diagram describing detection patterns according to an alternative embodiment.

FIG. 22 is a diagram describing detection patterns according to a further alternative embodiment.

FIG. 23 is a diagram describing combinations as respective detection patterns.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Preferred embodiments will now be described with reference to the accompanying drawings, wherein like reference numerals designate corresponding or identical elements throughout the various drawings. The drawings are to be viewed in an orientation in which the reference numerals are viewed correctly.

An embodiment of the present invention will be described with reference to the drawings.

5

FIG. 1 is a schematic overall side view of a working machine 1 according to the present embodiment. In this embodiment, the working machine 1 is exemplified by a backhoe serving as a kind of swivel working machine.

Referring to FIG. 1, the working machine 1 includes a machine body (or a swivel base) 2, a traveling device 3 and a working device 4. A cabin 5 is mounted on the machine body 2. An operator seat (or a seat) 6 on which a driver (or an operator) sits is provided in the cabin 5.

In the present embodiment, description will be given on the assumption that a forward direction corresponds to the forward direction (a direction designated by an arrow A1 in FIG. 1) from an operator sitting on the operator seat 6 in the working machine 1, a rearward direction corresponds to the rearward direction (a direction designated by an arrow A2 in FIG. 1) from the operator, a leftward direction corresponds to the leftward direction from the operator, and a rightward direction corresponds to the rightward direction from the operator. A fore-and-aft direction K1 is illustrated as the machine fore-and-aft direction, and a horizontal direction perpendicular to the fore-and-aft direction K1 is defined as a machine-width direction or a width direction of the machine body 2.

A machine-width distal direction or a distal direction in the machine-width direction corresponds to rightward or leftward direction from the central portion of the machine body 2. In other words, the machine-width distal direction is defined as the machine-width direction separating from the central portion of the machine body 2. A machine-width proximal direction or a proximal direction in the machine-width direction corresponds to the direction opposite to the machine-width distal direction. In other words, the machine-width proximal direction is defined as the machine-width direction approaching the central portion of the machine body 2.

Referring to FIG. 1, the traveling device 3 supports the machine body 2 travelably. The traveling device 3 includes a traveling frame 3A, a first traveling unit 3L provided on the left side of the traveling frame 3A, and a second traveling unit 3R provided on the right side of the traveling frame 3A. The first and second traveling units 3L and 3R are crawler traveling units. The first traveling unit 3L is driven by a first traveling motor M1. The second traveling unit 3R is driven by a second traveling motor M2. The first and second traveling motors M1 and M2 are hydraulic motors (or hydraulic actuators), for example.

A dozer 7 is attached to a front portion of the traveling device 3. The dozer 7 can be raised and lowered, i.e., a blade thereof can be moved upward and downward, by telescoping a dozer cylinder (or a hydraulic actuator).

Referring to FIG. 1, the machine body 2 is supported on the traveling frame 3A via a swivel bearing 8 so as to be swivelable around a swivel axis X1 that is a vertical center axis of the swivel bearing 8. A prime mover is mounted on the machine body 2. The prime mover is a diesel engine. Alternatively, it may be a gasoline engine, an LPG engine, or an electric motor. Alternatively, it may be a hybrid system including an engine and an electric motor.

Referring to FIG. 1, the machine body 2 includes a platform (hereinafter referred to as a swivel platform) 9 that is swivelable around the swivel axis X1. The swivel platform 9 is formed of a steel plate or so on such as to define a bottom of the machine body 2. The swivel platform 9 is provided on an upper face thereof close to the central portion thereof with longitudinal ribs 9L and 9R serving as reinforcement members extended between front and rear ends

6

thereof. The machine body 2 is provided at a rear portion thereof with a weight 10 erected on the swivel platform 9.

The machine body 2 is provided on a front portion thereof with a support unit 20 for supporting the working device 4. The support unit 20 includes a support bracket 20A and a swing bracket 20B. The support bracket 20A is fixed to front portions of the longitudinal ribs 9L and 9R and protrudes forward from the machine body 2. The swing bracket 20B is attached to a front portion of the support bracket 20A, which is the portion protruding from the machine body 2, via a swing shaft 26 so as to be swingable around a vertical axis. Accordingly, the swing bracket 20B is rotatable in the machine-width direction, i.e., horizontally around the swing shaft 26.

Referring to FIG. 1, the working device 4 is attached to the swing bracket 20B. The working device 4 includes a boom 22, an arm (serving as a support member) 23 and a working tool (serving as a supported member) 24. A base portion 22A of the boom 22 is pivotally supported by an upper portion of the swing bracket 20B via a boom shaft 27. The boom shaft 27 has a horizontal axis perpendicular to the vertical direction. Accordingly, the boom 22, when directed forward of the machine body 2, is rotatable around a lateral axis, i.e., an axis extended in the machine-width direction. The boom 22 swings upward or downward when it rotates around the boom shaft 27. The boom 22 is bent so as to be rearwardly convex at the longitudinally central portion thereof when it is raised at a highest position such as shown in FIG. 2.

A basal end portion (a second end portion other than a later-discussed first end portion) 23A of the arm 23 is pivotally supported by a tip end portion 22B of the boom 22 via an arm shaft 28. The arm shaft 28 has an axis parallel to the axis of the boom shaft 27. Accordingly, the arm 23 is rotatable around a lateral axis when the boom 22 is directed forward of the machine body 2. The arm 23 swings in a crowd direction defined as a direction approaching the boom 22 or in a dump direction defined as a direction away from the boom 22 when it rotates around the arm shaft 28.

In the present embodiment, the working tool 24 is exemplified by a bucket serving as a standard attachment attached to the working device 4. Hereinafter, the working tool 24 may be referred to as the bucket.

A base portion 24A of the working tool 24 is pivotally supported by a tip end portion (a first end portion) 23B of the arm 23 via a working tool shaft (serving as a pivot) 29. The working tool shaft 29 may be referred to as a bucket shaft. The working tool shaft 29 has an axis parallel to the axis of the arm shaft 28. Accordingly, the working tool 24 is rotatable around a lateral axis when the boom 22 is directed forward of the machine body 2. A tip end portion 24B of the working tool 24 swings in a crowd direction, i.e., a direction approaching the arm 23, or in a dump direction, i.e., a direction away from the arm 23, when it rotates around the arm shaft 28. In other words, the bucket 24 is movable for scooping and dumping. The scooping movement means the swing of the bucket 24 in the direction approaching the boom 22, i.e., the crowd direction, when scooping up earth or so on, for example. The dumping movement means the swing of the bucket 24 in the direction away from the boom 22, i.e., the dump direction, when dumping or discharging the scooped earth or so on, for example.

The working tool 24 is connected to the arm 23 via a linkage 30. The linkage 30 includes a first link 30A and a second link 30B. The first link 30A is pivotally supported at a first end thereof by the arm 23 via a first link shaft 31. The

7

second link 30B is pivotally supported at a first end thereof by the base portion 24A of the working tool 24 via a second link shaft 32. The first link 30A and the second link 30B are pivotally connected at second ends thereof to each other via a connection shaft 33. The first link shaft 31, the second link shaft 32 and the connection shaft 33 have respective axes parallel to the axis of the working tool shaft 29.

Another working tool driven by a hydraulic actuator, i.e., a hydraulic attachment, may be attached to the working machine 1 instead of or in addition to the bucket 24. Such another working tool is exemplified by a hydraulic breaker, a hydraulic crusher, an angle broom, an earth auger, a pallet fork, a sweeper, a mower, a snowblower or so on.

The swing bracket 20B is swingable by telescoping a swing cylinder C2 in the machine body 2. The boom 22 is swingable by telescoping a boom cylinder C3. The arm 23 is swingable by telescoping an arm cylinder C4. The working tool 24 is swingable by telescoping a working tool cylinder (a bucket cylinder) C5. The swing cylinder C2, the boom cylinder C3, the arm cylinder C4 and the working tool cylinder C5 are each constituted by a double acting hydraulic cylinder (hydraulic actuator).

Referring to FIG. 1, the working tool cylinder C5 is disposed forward from the arm 23. The working tool cylinder C5 is disposed along the arm 23 and is pivotally supported at a first end portion thereof by the basal end portion 23A of the arm 23. Specifically, the first end portion of the working tool cylinder C5 is pivotally supported via a cylinder shaft 35 by a cylinder bracket 34 fixed to the basal end portion 23A of the arm 23. The cylinder shaft 35 has an axis parallel to the axis of the arm shaft 28. The working tool cylinder C5 is pivotally supported at a second end portion thereof by the tip end portion 23B of the arm 23. Specifically, the second end portion of the working tool cylinder C5 is pivotally connected to the second end portions of the first and second links 30A and 30B via the connection shaft 33.

Referring to FIG. 2, the working tool cylinder C5 includes a cylinder tube 36 and a piston rod 37 which is extendable and contractible with respect to the cylinder tube 36, thereby being telescopically movable. Specifically, the working tool cylinder C5 includes a piston 38 axially movably incorporated in the cylinder tube 36. The piston rod 37 is joined to the piston 38 so that, when the piston 38 moves, the piston rod 37 axially moves together with the piston 38 so as to cause the extension or contraction movement of the working tool cylinder C5.

Referring to FIG. 2, the piston 38 divides the interior of the cylinder tube 36 into a first pressure chamber 36A on the bottom side and a second pressure chamber 36B on the rod side. The bottom side of the cylinder tube 36 is defined as the side through which the piston rod 37 is not passed. The rod side of the cylinder tube 36 is defined as the side opposite to the bottom side, i.e., the side through which the piston rod 37 is passed.

The piston rod 37 is provided with a rod head 37A at a tip end portion thereof opposite to the portion thereof joined to the piston 38. The rod head 37A is pivotally supported by the cylinder bracket 34 via the cylinder shaft 35. The cylinder tube 36 is provided with an attachment portion 36C at the bottom side end portion thereof. The attachment portion 36C is pivotally connected to the second end portions of the first and second links 30A and 30B via the connection shaft 33.

Referring to FIG. 3, when the working tool cylinder C5 is telescoped, the working tool 24 swings in a dump direction Y1 or a crowd direction Y2 around the working tool shaft (bucket shaft) 29. The working tool 24 swings around the working tool shaft 29 so that the tip end portion 24B swings

8

between a dump position (i.e., an end position on the dump side) P1 and a crowd position (i.e., an end position on the crowd side) P2. The dump position P1 is defined as the position of the working tool cylinder C5 when it is fully contracted (when in the fully contracted state). The crowd position P2 is defined as the position of the working tool cylinder C5 when it is fully extended (when in the fully extended state).

Referring to FIG. 2, the working tool cylinder C5 is constituted by a hydraulic cylinder telescopically moved by hydraulic fluid flowing through fluid passages, including a first fluid passage 39A and a second fluid passage 39B, formed in the piston rod 37. In other words, the working tool cylinder C5 is constituted by a pipeless hydraulic cylinder having no hydraulic fluid pipe on the outside of the cylinder tube 36. The first fluid passage 39A is fluidly connected to the first pressure chamber 36A, and the second fluid passage 39B is fluidly connected to the second pressure chamber 36B. The rod head 37A is formed therein with a first port 40A joined to the first fluid passage 39A, and with a second port 40B joined to the second fluid passage 39B (see FIG. 4). Hydraulic fluid hoses are coupled to the first port 40A and the second port 40B, respectively, so that hydraulic fluid is supplied or discharged to and from each of the first and second ports 40A and 40B via each of the hydraulic fluid hoses. The fluid supply to the first port 40A causes the extension movement of the working tool cylinder C5. The fluid supply to the second port 40B causes the contraction movement of the working tool cylinder C5.

Referring to FIGS. 2 and 3, the working tool cylinder C5 and the arm 23 are provided therebetween with a detection device (or a position sensor) 41 for detecting a swing position of the working tool 24. The swing position is defined as a position of the working tool 24 swinging around the working tool shaft 29. This position may correspond to the dump position P1, the crowd position P2, or any position between the dumping position P1 and the crowd position P2. Due to the pipeless hydraulic cylinder employed as the working tool cylinder C5, the working tool cylinder C5 and the arm 23 have a space without a pipe or a hose therebetween. This space is useful for arrangement of the detection device 41. Due to the arrangement of the detection device 41 between the working tool cylinder C5 and the arm 23, the detection device 41 is prevented from being damaged.

The detection device 41 detects a telescopic movement state (or a stroke length) of the working tool cylinder C5, thereby detecting the swing position of the working tool 24. Specifically, referring to FIGS. 2 and 3, the detection device 41 includes a detected object 42 provided on the cylinder tube 36 and a detector 43 provided on the piston rod 37. The detector 43, while moving together with the piston rod 37, detects the detected object 42, thereby detecting the telescopic movement state of the working tool cylinder C5. Alternatively, the detected object 42 may be provided on the piston rod 37 and the detector 43 may be provided on the cylinder tube 36.

Referring to FIG. 2, the detected object 42 is attached to an outer surface of the cylinder tube 36 facing the arm 23. Specifically, the detected object 42 is fastened with screws or so on to an attachment plate 44 fixed to the cylinder tube 36. The attachment plate 44 is disposed on the rod-side portion of the cylinder tube 36 and is fixed to the outer surface of the cylinder tube 36 facing the arm 23 by welding or so on.

Referring to FIGS. 5 to 7, guide members including a first guide member 45A and a second guide member 45B are provided on the attachment plate 44. The attachment plate

44 has a first fitting portion 44A which protrudes leftward and is fittingly covered with the first guide member 45A. The attachment plate 44 has a second fitting portion 44B which protrudes rightward and is fittingly covered with the second guide member 45B. The first fitting portion 44A and the second fitting portion 44B are disposed close to a first end of the attachment plate 44 (i.e., an end of the attachment plate 44 on the rod-side of the cylinder tube 36).

Referring to FIGS. 6 and 7, the detected object 42 includes a first detected object 42A and a second detected object 42B. The first detected object 42A is disposed on a left portion of the attachment plate 44, and the second detected object 42B is disposed on a right portion of the attachment plate 44. The first detected object 42A has a predetermined length in the longitudinal direction of the cylinder tube 36 such as to have a first end corresponding to the first end of the attachment plate 44 and a second end corresponding to a second end of the attachment plate 44 (i.e., another end of the attachment plate 44 on the bottom-side of the cylinder tube 36). The first detected object 42A includes a casing 46A and a plurality of magnets 47A. The casing 46A is formed to have the length corresponding to the whole length of the attachment plate 44 between the first and second ends of the attachment plate 44, and is fastened to the attachment plate 44 by screws or so on. The casing 46A may be made of a single member extending from a first end thereof to a second end thereof, or may be dividable into parts in the longitudinal direction thereof.

The magnets 47A are provided inside of the casing 46A so that they are aligned at intervals from the first end of the casing 46A to the second end of the casing 46A. The magnets 47A may be replaced with a continuously formed magnet. The second detected object 42B is provided on a portion of the attachment plate 44 close to the first end of the attachment plate 44. The second detected object 42B includes a casing 46B and a single magnet 47B. The second detected object 42B may be omitted. To enhance the reliability of detection device 41, the second detected object 42B may be formed to have the same length as that of the first detected object 42A, and may be disposed to positionally coincide to the first detected object 42A in the longitudinal direction of the cylinder tube 36.

Referring to FIGS. 6 and 7, the detector 43 includes a first sensor 43A and a second sensor 43B. The first sensor 43A and the second sensor 43B function as proximity sensors and as magnetic sensors for detecting magnetism. The first sensor 43A and the second sensor 43B are disposed to have a space therebetween in the machine width direction and are disposed to positionally coincide to each other in the longitudinal direction of the cylinder tube 36.

The first sensor 43A corresponds to the first detected object 42A so as to detect the first detected object 42A. Specifically, the first sensor 43A includes a base member 48A movable together with the piston rod 37, and a detection element 49A for detecting the magnets 47A. While the base member 48A moves together with the piston rod 37, the detection element 49A scans the magnets 47A one after another, thereby detecting any portion of the first detected object 42A between the first and second ends of the first detected object 42A.

The second sensor 43B corresponds to the second detected object 42B so as to detect the second detected object 42B. Specifically, the second sensor 43B includes a base member 48B movable together with the piston rod 37, and a detection element 49B for detecting the magnet 47B. Due to the movement of the base member 48B together with the piston rod 37, the detection element 49B detects a

portion of the second detected object 42B between first and second ends of the second detected object 42B.

FIGS. 5 and 6 and the lowest drawing of drawings in FIG. 9 illustrate the working tool cylinder C5 in the fully contracted state. In this state, the working tool 24 is disposed at the dump position P1 (see FIG. 3). In the fully contracted state of the working tool cylinder C5, the detector 43 is disposed at a first position P4 corresponding to the second end of the first detected object 42A. At this time, the first sensor 43A detects the first detected object 42A while the second sensor 43B does not detect the second detected object 42B. When the working tool cylinder C5 extends from the fully contracted state, the detector 43 moves together with the piston rod 37. Accordingly, the detector 43 keeps detecting the first detected object 42A until the detector 43 reaches a second position P5 corresponding to the first end of the first detected object 42A as illustrated by the middle drawing of FIG. 9. On the other hand, the second sensor 43B detects the second detected object 42B while the detector 43 moves from a position slightly before the second position P5 to the second position P5. The position of the working tool 24, when the detector 43 is disposed at the second position P5, is referred to as a first predetermined position P3 (see FIG. 3).

At the second position P5 and in the vicinity thereof, the first sensor 43A detects the first detected object 42A and the second sensor 43B detects the second detected object 42B, thereby enhancing the reliability of the detection device 41 when at the second position P5 and in the vicinity thereof.

After the detected object 42 passes the second position P5 by further extension movement of the working tool cylinder C5 from the state illustrated by the middle drawing in FIG. 9, the detector 43 no further detects the detected object 42 until the working tool cylinder C5 is fully extended as illustrated by the highest drawing in FIG. 9.

Referring to FIG. 3, the first predetermined position P3 is set between the dump position P1 and the crowd position P2, and the detection device 41 detects the existence of the working tool 24 when it is in a predetermined range E1 between the first predetermined position P3 and the dump position P1. Specifically, the predetermined range E1 ranges from the first predetermined position P3 to the dump position P1.

Referring to FIG. 3, a reference numeral T1 designates a movement locus of the tip end portion 24B of the working tool 24 swinging from the dump position P1 to the crowd position P2, and a reference numeral O1 designates a middle point of the movement locus T1. The movement range of the tip end portion 24B from the middle point O1 to the dump position P1 is defined as a dump side in the swing range of the working tool 24, and the movement range of the tip end portion 24B from the middle point O1 to the crowd position P2 is defined as a crowd side in the swing range of the working tool 24. On this assumption, the detection device 42 detects the existence of the working tool 24 when it is in the predetermined range E1 on the dump side in the swing range thereof.

An attachment structure 51 configured to attach the detector 43 to the piston rod 37 will now be described.

Referring to FIGS. 5 to 7, the attachment structure 51 includes a detector mount 52, to which the detector 43 is attached, and a connection mechanism 53 to connect the detector mount 52 to the piston rod 37.

The detector mount 52 includes a mount main body 54, a supporter 55 for supporting the mount main body 54 on the attachment plate 44, and a cover plate 56 for covering the detected object 42.

11

The mount main body **54** includes a base wall **54a**, a first side wall **54b**, a second side wall **54c**, a first end wall **54d** and a second end wall **54e**. The base wall **54a** is disposed between the working tool cylinder **C5** and the arm **23**. The first side wall **54b** extends from a left end of the base wall **54a**. The second side wall **54c** extends from a right end of the base wall **54a**. The first end wall **54d** is provided on a first end portion of the base wall **54a**, i.e., an end portion of the base wall **54a** on the rod-side of the cylinder tube **36**. The second end wall **54e** is provided on a second end portion of the base wall **54a**, i.e., another end portion of the base wall **54a** on the bottom-side of the cylinder tube **36**.

Referring to FIG. 5, the base wall **54a** faces the detected object **42** when the working tool cylinder **C5** is fully contracted. An attachment block **57** is fixed to the second end portion of the base wall **54a**. The first sensor **43A** is attached to the left side of the attachment block **57**. The second sensor **43B** is attached to the right side of the attachment block **57**.

Referring to FIG. 7, the supporter **55** includes a first slide member **55A** on the left side thereof, and includes a second slide member **55B** on the right side thereof. The first slide member **55A** is formed to have a rightward open groove into which the first fitting portion **44A** is fitted so that the first slide member **55A** is movable relative to the attachment plate **44** in the longitudinal direction of the cylinder tube **36**. The second slide member **55B** is formed to have a leftward open groove into which the second fitting portion **44B** is fitted so that the second slide member **55B** is movable relative to the attachment plate **44** in the longitudinal direction of the cylinder tube **36**. The first slide member **55A** is fixed to the first side wall **54b**, and the second slide member **55B** is fixed to the second side wall **54c**. Accordingly, the attachment block **57** is supported by the attachment plate **44** via the supporter **55** movably relative to the attachment plate **44** in the longitudinal direction of the cylinder tube **36**.

Referring to FIG. 5, the supporter **55** protrudes from the mount main body **54** to the bottom side and to the rod side. The portion of the supporter **55** protruding to the bottom side from the mount main body **54** extends from the detector mount **52** to the vicinity of a cover member **58** covering a grease nipple as shown in the lowest drawing in FIG. 9.

Referring to FIG. 8, the cover plate **56** is disposed on the side of the attachment plate **44** opposite to the working tool cylinder **C5** so as to connect the first slide member **55A** and the second slide member **55B** to each other. Referring to FIG. 5, the cover plate **56** is fixed at a first end thereof to the second end wall **54e** of the mount main body **54**.

Referring to FIG. 9, the cover plate **56** is extended to have a second end defining an end portion of the supporter **55**. When the working tool cylinder **C5** is extended, the cover plate **56** moves together with the piston rod **37** so as to cover the detected object **42**.

Referring to FIGS. 5 and 6, the connection mechanism **53** includes a first connection piece **59**, a second connection piece **60** and a connection pin **61**. The first connection piece **59** is joined to the piston rod **37**. Specifically, the first connection piece **59** is fixed to a fixture plate **62** attached to the rod head **37A**. The second connection piece **60** is joined to the mount main body **54** (of the detector mount **52**). Specifically, the second connection piece **60** includes a first portion **60a** and a second portion **60b**. The first portion **60a** is fixed to the first end wall **54d** of the mount main body **54** so as to connect the first slide member **55A** and the second slide member **55B** to each other. The second portion **60b** extends from the first portion **60a** toward the rod head **37A**. The connection pin **61** is passed through the first connection

12

piece **59** and the second connection piece **60** so as to couple the first and second connection pieces **59** and **60** to each other. Specifically, referring to FIG. 5, the first connection piece **59** and the second portion **60b** of the second connection piece **60** face each other in the radial direction of the piston rod **37** so as to have a space **63** therebetween, and the connection pin **61** is passed through the space **63**.

Referring to FIG. 10, the first connection piece **59** is formed with a first pin hole **64** through which the connection pin **61** is passed. The first pin hole **64** is formed into a long hole shape such as to allow the piston rod **37** to rotate around an axis **B1** of the piston rod **37**. Specifically, an extension line **L1** of the axis of the connection pin **61** is perpendicular to the axis of the piston rod **37**. The long hole serving as the first pin hole **64** elongates parallel to a direction **L2** perpendicular to the extension line **L1** and the axis **B1**.

The second portion **60b** of the second connection piece **60** is formed with a second pin hole **65** through which the connection pin **61** is passed. The second pin hole **65** is formed into a circular hole shape.

Due to backlash of the cylinder shaft **35** in a cylinder shaft passage hole penetrating the rod head **37A** and the cylinder bracket **34**, the piston rod **37** is allowed to slightly rotate around the axis **B1** thereof so that the first connection piece **59** is allowed to swing according to the rotation of the piston rod **37**.

FIGS. 11 to 14 illustrate an alternative embodiment.

In the alternative embodiment, referring to FIG. 11, a second predetermined position **P7** is set between a first predetermined position **P6** and the dump position **P1**, and the detection device **41** detects the existence of the working tool **24** when it is in a predetermined range **E2** ranging from the first predetermined position **P6** to the second predetermined position **P7**. In the alternative embodiment, the first predetermined position **P6** is defined as a position where an angle **D3** between a line **L3** connecting the axial center of the cylinder shaft **35** to the axial center of the connection shaft **33** and a line **L4** connecting the axial center of the first link shaft **31** to the axial center of the connection shaft **33** becomes a substantially right angle.

Referring to FIGS. 12 and 13 and the lowest drawing of drawings in FIG. 14, when the working tool cylinder **C5** is fully contracted, the detector **43** is separated from the detection device **42** so that the detection device **41** does not detect the working tool **24**. While the working tool cylinder **C5** is extended from the fully contracted state, the detection device **41** does not detect the working tool **24** before the first sensor **43A** reaches a position corresponding to the second end of the first detected object **42A**.

When the first sensor **43A** located at the position corresponding to the second end of the first detected object **42A** detects the first detected object **42A**, the detection device **41** detects the existence of the working tool **24** at the second predetermined position **P7**. While the working tool cylinder **C5** is extended further from this extension state, the working tool **24** is detected until the detector **43** reaches a position corresponding to the first end of the detected object **42**. When the detector **43** reaches the position corresponding to the first end of the detected object **42** and detects the first detected object **42A** and the second detected object **42B**, the detection device **41** detects the existence of the working tool **24** at the first predetermined position **P6**. Accordingly, while the working tool **24** is in the predetermined range **E2** ranging from the first predetermined position **P6** to the second predetermined position **P7**, the detection device **41** detects the existence of the working tool **24** in the predetermined range **E2**.

13

In FIG. 14, lowest, middle and highest drawings indicate different telescopic movement states of the working tool cylinder C5. The lowest drawing illustrates the working tool cylinder C5 when it is fully contracted. The middle drawing illustrates the working tool cylinder C5 when the working tool 24 is located at the first predetermined position P6. The highest drawing illustrates the working tool cylinder C5 when it is fully extended.

In the alternative embodiment, as illustrated by the highest drawing in FIG. 14, the detected object 42 is covered with the cover plate 56 without protruding from the cover plate 56 even when the working tool cylinder C5 is fully extended.

All constitution elements of the alternative embodiment except for the above-mentioned elements are identical or similar to those of the foregoing embodiment shown in FIGS. 1 to 10.

Further alternative embodiments will now be described with reference to FIGS. 15 to 23.

In the embodiments shown in FIGS. 15 to 23, the detection device 41 and the attachment structure 51 are identical or similar to those shown in FIGS. 7, 8, 12 and 13. Illustration and description of the structures identical or similar to those of the foregoing embodiments will be omitted.

In the present embodiments, referring to FIG. 17, the working machine 1 includes a controller 71 for controlling the swing of the bucket (working tool) 24, and includes a bucket control valve 72 for controlling the bucket cylinder C5. The controller 71 is constituted by a microcomputer including CPU (Central Processing Unit) and EEPROM (Electrically Erasable Programmable Read-Only Memory), for example.

The bucket control valve 72 is a control valve electrically controlled by the controller 71. For example, a proportional directional solenoid valve serves as the bucket control valve 72. The proportional directional solenoid valve is a valve in which a main spool is moved by a solenoid to control the flow of hydraulic fluid. The bucket control valve 72 is a three position shift valve shiftable among a neutral position 72a, a first position 72b and a second position 72c. The bucket control valve 72 includes a first solenoid 72d and a second solenoid 72e. The first solenoid 72d and the second solenoid 72e are electrically connected to the controller 71 so that each of the first and second solenoids 72d and 72e is excited or unexcited according to a command signal outputted from the controller 71. Due to the excitation or non-excitation of each of the first and second solenoids 72d and 72e, the bucket control valve 72 is shiftable from the neutral position 72a to either the first position 72b or the second position 72c.

The bucket control valve 72 is fluidly connected to a hydraulic pump 92 via a supply fluid passage 73A, and to a tank 74 via a drain fluid passage 73B. The bucket control valve 72 is fluidly connected to the piston rod 37 of the bucket cylinder C5 via a first cylinder fluid passage 73C and a second cylinder fluid passage 73D. Specifically, the first cylinder fluid passage 73C is fluidly connected to the first fluid passage 39A, and the second cylinder fluid passage 73D is fluidly connected to the second fluid passage 39B.

Referring to FIG. 17, an operation member 75 for operating the bucket 24 is electrically connected to the controller 71. The controller 71 can obtain an operation signal (electric signal) from the operation member 75. The operation member 75 is disposed adjacent to the operator seat 6 and includes a lever 76 which can be gripped and operated by an operator. The lever is swingable from its neutral position in first and second directions opposite to each other. For

14

example, when the level 76 is swung in the first direction, the first solenoid 72d is excited to shift the bucket control valve 72 to the first position 72b. Due to the shift of the bucket control valve 72 to the first position 72b, the bucket cylinder C5 is contracted to swing the bucket 24 in the dump direction Y1. When the level 76 is swung in the second direction, the second solenoid 72e is excited to shift the bucket control valve 72 to the second position 72c. Due to the shift of the bucket control valve 72 to the second position 72c, the bucket cylinder C5 is extended to swing the bucket 24 in the crowd direction Y2. When the lever 76 is returned to its neutral position, the bucket control valve 72 returns to the neutral position 72a to stop the telescopic movement of the bucket cylinder C5, i.e., to stop the movement of the bucket 24.

Referring to FIG. 18, the bucket cylinder C5, when in a fully contracted state 77, is disposed parallel and adjacent to the arm 23. Assuming that the bucket cylinder C5 is extended from the fully contracted state 77 to a fully extended state 78, in a first stage of the extension movement of the bucket cylinder C5, the bucket cylinder C5 swings away from the arm 23 so as to increase a swing angle G of the bucket cylinder C5. After passing a halfway point of the extension movement of the bucket cylinder C5, the swing direction of the bucket cylinder C5 relative to the arm 23 is reversed so that the bucket cylinder C5 approaches the arm 23 so as to gradually reduce the swing angle G of the bucket cylinder C5. In FIG. 18, a reference numeral 79 designates a turning point as the halfway point of extension or contraction movement of the bucket cylinder C5, where the variation direction of the swing angle G of the bucket cylinder C5 is reversed between the increasing direction and the reducing direction, i.e., where the swing angle G becomes maximum. In FIG. 16, a reference numeral P6 designates a position of the bucket 24 when the bucket cylinder C5 is at the turning point 79. The following description is given on an assumption that the state where the bucket cylinder C5 is disposed at the turning point 79 and the bucket 24 is disposed at the position P6 is referred to as a neutral position 80. In other words, the neutral position 80 is conceptual. Referring to FIG. 16, the neutral position 80 corresponding to the turning point 79 of the bucket cylinder C5 defines the border between the dump side E3 and the crowd side E4 in the swing range of the bucket 24.

Referring to FIG. 18, an angle sensor 81 is attached to the cylinder bracket 34 so as to detect the swing angle G of the bucket cylinder C5 around the cylinder shaft 35. The angle sensor 81 is constituted by a potentiometer, for example. The angle sensor 81 detects a swing angle G1 on the dump side E3 from the neutral position 80 and a swing angle G2 on the crowd side E4 from the neutral position 80. The angle sensor 81 is linked to the rod head 37A of the bucket cylinder C5 by a linkage 82. Accordingly, the angle sensor 81 with the linkage 82 detects the rotation of the rod head 37A around the cylinder shaft 35, thereby detecting the swing angle G of the bucket cylinder C5 around the cylinder shaft 35. Alternatively, the angle sensor 81 may have no linkage to directly detect the rotation of the bucket cylinder C5 around the cylinder shaft 35.

Referring to FIG. 17, the angle sensor 81 is electrically connected to the controller 71. The controller 71 can obtain a detection value (or a potentiometer resistance value) from the angle sensor 81. The controller 71 includes a calculation unit 83. The calculation unit 83 calculates the swing position of the bucket 24 based on the swing angle G of the bucket cylinder C5, i.e., the potentiometer resistance value. The

15

swing position of the bucket **24** is defined as any position of the bucket **24** swinging around the bucket shaft **29**.

In the above-mentioned operation mechanism of the bucket **24**, because of the reversing of variation direction of the swing angle **G** of the bucket cylinder **C5** at the halfway point of the telescopic movement thereof, it is not clear whether the swing position of the bucket **24** corresponding to the detected swing angle **G** is on either the dump side **E3** or the crowd side **E4** with respect to the neutral position **80**.

Therefore, referring to FIG. **16**, the detection device **41** is provided to detect whether the bucket **24** is on the dump side **E3** or the crowd side **E4** with respect to the neutral position **80**. In other words, the calculation unit **83** of the controller **71** calculates or determines the swing position of the bucket **24** based on the swing angle **G** detected by means of the angle sensor **81** and detection information as a result of detection by means of the detection device **41**.

The detection device **41** detects ON and OFF signals such as to detect the position of the piston rod **37** relative to the cylinder tube **36** during the telescopic movement of the bucket cylinder **C5**. The detector **43**, when detecting the detected object **42**, outputs a detection signal to the controller **71**. The detection signal may be either the on signal or the off signal.

FIGS. **12** and **13** and the lowest drawing of drawings in FIG. **19** illustrate the bucket cylinder **C5** when in the fully contracted state **77**. When the bucket cylinder **C5** is in the fully contracted state **77**, the detector **43** is located at the first position **P4** corresponding to the second end of the first detected object **42A**. The detector **43** is separated away from the detected object **42** so that the first sensor **43A** does not detect the first detected object **42A** and the second sensor **43B** does not detect the second detected object **42B**.

When the bucket cylinder **C5** is extended from the fully contracted state **77**, the detector **43** moves together with the piston rod **37**. At the first stage of the movement of the detector **43**, the first sensor **43A** detects the first detected object **43A**. Referring to the middle drawing of the drawings in FIG. **19**, when the detector **43** reaches the second position **P5** corresponding to the first end of the first detected object **42A**, the second sensor **43B** detects the second detected object **42B**. The state where the first sensor **43A** detects the first detected object **42A** and the second sensor **43B** detects the second detected object **42B** is defined as the neutral position **80**. The detector **43** detects the neutral position **80** by detecting the detectable end of the second detected object **42B** of the detected object **42**.

As illustrated, the detection of the neutral position **80** requires both the detection of the first detected object **42A** by means of the first sensor **43A** and the detection of the second detected object **42B** by means of the second sensor **43B**, thereby enhancing the reliability of the detection device **41**. Alternatively, the second detected object **42B** and the second sensor **43B** may be omitted.

While the bucket cylinder **C5** is further extended, after the detector **43** passes the second position **P5**, the detector **43** does not detect the detected object **42** until the bucket cylinder **C5** reaches a third position **P8** where it is in the fully extended state **78** as illustrated by the highest drawing of the drawings in FIG. **19**.

Referring to FIG. **17**, the detection device **41** is electrically connected to the controller **71**. The controller **71** can obtain detection information from the detection device **41**. The controller **71** includes a determination unit **84** such that, based on the detection information from the detection device **41**, the determination unit **84** determines whether the bucket **24** exists on the dump side **E3** or the crowd side **E4**

16

with respect to the neutral position **80**. The working machine **1** includes a switch **85**, serving as an instruction input unit, for receiving an instruction input from the operator, the instruction input indicating whether the bucket **24** is on the dump side **E3** or the crowd side **E4**. The switch **85** is disposed adjacent to the operator seat **6**. The switch **85** is electrically connected to the controller **71**. The controller **71** can obtain an electric signal from the switch **85**. The controller **71** includes a storage unit **86**.

Referring to FIG. **20**, the detection device **41** is arranged so that it can find whether the bucket **24**, when existing in a detectable range that is a predetermined range as the vicinity of the neutral position **80**, is on the dump side **E3** or the crowd side **E4**. In other words, the detection device **41** serves as a sensor for judging whether the bucket **24** exists on the dump side **E3** or the crowd side **E4** with respect to the neutral position **80**. In this embodiment, while the angle sensor **81** alone is insufficient to detect the swing position of the bucket **24** in the vicinity of the neutral position **80** at which the variation direction of the potentiometer resistance value is reversed, the detection information from the detection device **41** is used to determine the swing position of the bucket **24**.

Accordingly, the determination of whether the bucket **24** is on the dump side **E3** or the crowd side **E4** is not performed when the bucket **24** is out of the predetermined range as the vicinity of the neutral position **80**. Therefore, at the start of processing to detect the swing angle of the bucket **24**, a position confirmation operation, i.e., an initial position determining operation, to determine whether the bucket **24** exists on the dump side **E3** or the crowd side **E4** is performed while the bucket **24** is out of the predetermined range as the vicinity of the neutral position **80**. For example, the position confirmation operation is performed in the following way.

First, the operator temporarily moves the bucket **24** to the dump side **E3** or the crowd side **E4**. When the bucket **24**, swung to the dump side **E3**, reaches the dump position **P1**, the operator pushes the switch **85** to inform the controller **71** that the bucket **24** is disposed at the dump position **P1** on the dump side **E3**. When the bucket **24**, swung to the crowd side **E4**, reaches the crowd position **P2**, the operator pushes the switch **85** to inform the controller **71** that the bucket **24** is disposed at the crowd position **P2** on the crowd side **E4**. The controller **71**, when informed of the position of the bucket **24**, makes the storage unit **86** store the informed position of the bucket **24**.

Alternatively, the controller **71** may automatically recognize whether the bucket **24** exists on the dump side **E3** or the crowd side **E4**. Specifically, after the bucket **24**, temporarily swung to the dump side **E3** or the crowd side **E4**, reaches the dump position **P1** or the crowd position **P2**, the resistance value of the potentiometer serving as the angle sensor **81** varies no further from that corresponding to the bucket **24** at the dump position **P1** or the crowd position **P2**. Therefore, if the state where the resistance value no further varies continues for a predetermined period, the controller **71** may automatically detect that the bucket **24** is disposed at the dump position **P1** or the crowd position **P2**. The controller **71**, when automatically recognizing the position of the bucket **24**, makes the storage unit **86** store the recognized position of the bucket **24**.

Description will now be given of the determination of whether the bucket **24** in the vicinity of the neutral position **80** exists on the dump side **E3** or the crowd side **E4**.

Referring to FIG. **20**, in this embodiment, a detectable range **87** for the detector **43** to detect the detected object **42** ranges from the neutral position **80** to an intermediate

17

position between the neutral position **80** and the end position on the dump side **E3**. For example, the detection device **41**, when in the detectable range **87**, is turned on, while the detection device **41**, when out of the detectable range **87**, i.e., when in a first undetectable range **88** or a second undetectable range **89**, is turned off. In other words, the neutral position **80** is defined as a border between the turn-on range and the turn-off range. The voltage of the angle sensor **81**, when detecting the swing angle of the bucket cylinder **C5** in the vicinity of the neutral position **80**, is known. Therefore, if the angle sensor **81** is turned on while the voltage of the angle sensor **81** indicates that when detecting in the vicinity of the neutral position **80**, the determination unit **84** determines the bucket **24** as being on the dump side **E3**. If the angle sensor **81** is turned off while the voltage of the angle sensor **81** indicates that when detecting in the vicinity of the neutral position **80**, the determination unit **84** determines the bucket **24** as being on the crowd side **E4**.

It is assumed that the detector **43** is disposed in the first undetectable range **88** and then the bucket **24** (or the piston rod **37**) is moved in a first direction **D1** defined as a direction from the dump side **E3** to the crowd side **E4**. On this assumption, the detection device **41** is turned on as soon as the detector **43** enters the detectable range **87**. When the bucket **24** (or the piston rod **37**) is further moved in the first direction **D1** and the detector **43** leaves the detectable range **87** and enters the second undetectable range **89**, the detection device **41** is turned off. Accordingly, the determination unit **84** determines the bucket **24** as being on the crowd side **E4**. In this way, a first detection pattern appearing in the ON/OFF signal detection by means of the detection device **41** during the movement of the bucket **24** (or the piston rod **37**) in the first direction **D1** is indicated as “OFF-to-ON-to-OFF” (“Non-detection to Detection to Non-detection”). The result of determination by the determination unit **84** is stored in the storage unit **86**. After the bucket **24** in the vicinity of the neutral position **80** is determined as being on the crowd side **E4**, while the bucket **24** is swung further on the crowd side **E4** so as to move the detector **43** toward the crowd-side end position, the controller **71** keeps the determination, stored in the storage unit **86**, that the bucket **24** is on the crowd side **E4**.

It is assumed that the detector **43** is disposed in the second undetectable range **89** and then the bucket **24** (or the piston rod **37**) is moved in a second direction **D2** defined as a direction from the crowd side **E4** to the dump side **E3**. On this assumption, the controller **71** keeps the determination, stored in the storage unit **86**, that the bucket **24** is on the crowd side **E4** until the detector **43** reaches a point immediate before the neutral position **80**. When the detector **43** passes the neutral position **80**, the detection device **41** having been turned off is turned on so that the determination unit **84** determines the bucket **24** as being on the dump side **E3**. In this way, a second detection pattern appearing in the ON/OFF signal detection by means of the detection device **41** during the movement of the bucket **24** (or the piston rod **37**) in the second direction **D2** is indicated as “OFF-to-ON” (“Non-detection to Detection”). The controller **71** makes the storage unit **86** store the result of determination that the bucket **24** is on the dump side **E3**. Afterward, while the piston rod **37** is moved in the second direction **D2** and when the piston rod **37** is stopped with the detector **43** left in the first undetectable range **88**, the controller **71** keeps the state stored in the storage unit **86**, i.e., the state of the bucket **24** being on the dump side **E3**.

When the bucket **24** is moved again after the bucket **24** is stopped with the detector **43** left in the detectable range **87**,

18

the detector **43** leaves the detectable range **87** and enters either the first undetectable range **88** or the second undetectable range **89**, thereby causing “ON-to-OFF” pattern. For this reason, the detection device **41** alone is insufficient to determine whether the detector **43** enters the first undetectable range **88** or the second undetectable range **89**. To solve the problem, assuming that the detectable range **87** has a first detectable range end **87a** corresponding to the neutral position **80** and a second detectable range end **87b** opposite to the first detectable range end **87a**, when the detector **43** moves from the detectable range **87** to the first undetectable range **88** according to the movement of the piston rod **37** in the second direction **D2**, a potentiometer resistance value generated at the second detectable range end **87b** is considered to determine the detector **43** as entering the first undetectable range **88**. When the detector **43** moves from the detectable range **87** to the second undetectable range **89** according to the movement of the piston rod **37** in the first direction **D1**, another potentiometer resistance value generated at the first detectable range end **87a** corresponding to the neutral position **80** is considered to determine the detector **43** as entering the second undetectable range **89**.

Alternatively, the detection device **41**, when in the detectable range **87**, may be turned off, and the detection device **41**, when out of the detectable range **87**, may be turned on. In this case, the first detection pattern realized by means of the detection device **41** when the piston rod **37** is moved in the first direction **D1** is indicated as “ON-to-OFF-to-ON” (corresponding to “Non-detection to Detection to Non-detection”). The second detection pattern realized by means of the detection device **41** when the piston rod **37** is moved in the second direction **D2** is indicated as “ON-to-OFF” (corresponding to “Non-detection to Detection”) pattern.

Alternatively, the detectable range **87** for the detector **43** to detect the detected object **42** may range from the neutral position **80** to an intermediate position between the neutral position **80** and the end position on the crowd side **E4**.

As mentioned above, in this embodiment, the detection device **41** detects ON and OFF signals indicating whether the bucket cylinder **C5** is extended or contracted from the neutral position **80**. Each of the first detection pattern and the second detection pattern is a combination of ON-to-OFF and OFF-to-ON signal shifts, and the combination as the first detection pattern and the combination as the second detection pattern are different from each other.

Referring to FIG. 17, the working machine **1** includes a display unit (or a meter panel) **90**. The display unit **90** is disposed adjacent to the operator seat **6** visually from the operator. The display unit **90** is electrically connected to the controller **71**. The display unit **90** may be provided with the switch **85**. The controller **71** includes a command unit **91**.

When operation of the bucket **24** is ended, the controller **71** stores the position of the bucket **24** on either the dump side **E3** or the crowd side **E4**, and defines the stored position of the bucket **24** as an initial position for restart of the operation of the bucket **24**. When the bucket cylinder **C5** is telescoped from the initial position, i.e., when the piston rod **37** is moved from the initial position in either the first direction **D1** or the second direction **D2**, the controller **71** determines whether the bucket **24** is on the dump side **E3** or the crowd side **E4** with respect to the neutral position **80**. For example, the controller **71** makes the storage unit **86** store the position of the bucket **24** at the time immediate before a key switch is turned off. After the key switch is turned on again, the controller **71** views the position of the bucket **24** stored in the storage unit **86** so as to determine whether the bucket **24** is on the dump side **E3** or the crowd side **E4** with

19

respect to the neutral position **80**. It is greatly convenient because the operator does not have to perform the position confirmation operation after the key switch is turned on.

When the difference between the value of the angle sensor **81** before the key switch is turned off and that after the key switch is turned on is so great as to equal or exceed a hysteresis level, it is not sure that the bucket **24** exists at the same position as that before the key switch is turned off. In such a case, the controller **71** informs the operator of the uncertainty in position of the bucket, i.e., commands the display unit **90** to display a message for urging the operator to perform the position confirmation operation to determine whether the bucket **24** is on the dump side **E3** or the crowd side **E4**. Specifically, the command unit **91** commands the display unit **90** to display a message for urging the operator to move the bucket **24** to the dump side **E3** or the crowd side **E4**. Accordingly, the display unit **90** displays the message for urging the operator to locate the bucket **24** at any position on either the dump side **E3** or the crowd side **E4** and to register the position defined as the initial position. Therefore, the operator is only required to perform the position confirmation operation.

FIG. **21** illustrates a first modification of the embodiment shown in FIGS. **15** to **20**.

In the first modification shown in FIG. **21**, two detected objects **42**, i.e., detected objects **42L** and **42R**, are offset from each other in the longitudinal direction of the bucket cylinder **C5** so as to have a space therebetween where the neutral position **80** is set. The detected object **42L** is disposed to be detectable when the bucket **24** is on the dump side **E3**. The detected object **42R** is disposed to be detectable when the bucket **24** is on the crowd side **E4**. In other words, the neutral position **80** defines a border between two detectable ranges, i.e., a first detectable range **87A** and a second detectable range **87B**. An interval (or a distance) **W1** between the first detectable range **87A** and the neutral position **80** is different from an interval (or a distance) **W2** between the second detectable range **87B** and the neutral position **80**. The angle sensor **81** outputs a detection value (i.e., an angle sensor value) corresponding to the swing angle of the bucket cylinder **C5**. Therefore, based on the detection value of the angle sensor **81** at the time of the ON/OFF shift of the detection device **41**, the controller **71** determines whether the bucket **24** is on the dump side **E3** or the crowd side **E4** with respect to the neutral position **80**.

In the configuration shown in FIG. **21**, the detected objects **42L** and **42R** are illustrated to have the interval therebetween, however, the configuration is not limited to the illustrated configuration. The only requirement is that the detection value of the angle sensor **81** when the bucket **24** is at a position corresponding to one ON/OFF shift point in detection by means of the detection device **41** is different from that when the bucket **24** is at another position corresponding to the other ON/OFF shift point in detection by means of the detection device **41**.

The above-mentioned determination may be performed by using times **t1** and **t2**. The first detectable range **87A** has a detectable range end **87Aa** closer to the neutral position **80** than the other end thereof, and the time **t1** is defined as a time taken for movement of the detector **43** through the first detectable range **87A** from the detectable range end **87Aa** to the neutral position **80**. The second detectable range **87B** has a detectable range end **87Ba** closer to the neutral position **80** than the other end thereof, and the time **t2** is defined as a time taken for movement of the detector **43** through the second detectable range **87B** from the detectable range end **87Ba** to the neutral position **80**. The times **t1** and **t2** are calculated

20

based on the distances **W1** and **W2** from the neutral position **80** and the swing speed of the bucket **24**, and are inputted to the controller **71**.

In this case, for example, it is assumed that the detection device **41** is turned on when the detector **43** detects the detected object **42L**. On this assumption, the first detection pattern appearing in detection by means of the detection device **41** when the piston rod **37** is moved in the first direction **D1** is indicated as "OFF-to-ON-to-OFF" with the time **t1**. Based on this detection pattern, the bucket **24** can be determined as moving from the dump side **E3** to the crowd side **E4** and then being on the crowd side **E4**.

The second detection pattern appearing in detection by means of the detection device **41** when the piston rod **37** is moved in the second direction **D2** is indicated as "OFF-to-ON-to-OFF" with the time **t2**. Based on this detection pattern, the bucket **24** can be determined as moving from the crowd side **E4** to the dump side **E3** and then being on the dump side **E3**.

If the detected objects **42L** and **42R** have small distances from the neutral position **80**, i.e., if the times **t1** and **t2** are short, after one of the detected objects **42L** and **42R** is detected and before the other of the detected objects **42L** and **42R** is detected, the bucket **24** may be determined as being on the same side with the last detected detected object **42L** or **42R**. In the first modification, the detection device **41** may be turned off when the detector **43** detects either the detected object **42L** or **42R**.

As mentioned above, each of the first and second detection patterns may be a combination of ON-to-OFF and OFF-to-ON signal shifts with the turn-on time and the turn-off time, while the combination as the first detection pattern and the combination as the second detection pattern are different from each other.

FIG. **22** illustrates a second modification different from the first modification of FIG. **21**.

In the second modification shown in FIG. **22**, the detected object **42** includes a plurality of divisional members **42a** to **42c** which are offset from one another in the longitudinal direction of the bucket cylinder **C5** so as to have intervals thereamong.

In the second modification shown in FIG. **22**, the first detection pattern appearing in detection of ON and OFF signals by means of the detection device **41**, when the piston rod **37** is moved in the first direction **D1** from the dump side **E3** to the crowd side **E4**, is indicated as "OFF-to-ON-to-OFF-to-ON-to-OFF to-ON-to-OFF." Based on this detection pattern, the bucket **24** can be determined as moving from the dump side **E3** to the crowd side **E4** and then being on the crowd side **E4**.

The second detection pattern appearing in detection of ON and OFF signals by means of the detection device **41**, when the piston rod **37** is moved in the second direction **D2**, is indicated as "OFF-to-ON." Based on this detection pattern, the bucket **24** can be determined as moving from the crowd side **E4** to the dump side **E3** and then being on the dump side **E3**.

Constitution elements of the first and second modifications except for those mentioned above are identical or similar to those of the embodiment shown in FIGS. **15** to **20**.

As mentioned above, as an alternative embodiment, each of the first and second detection patterns may be a combination of the ON-to-OFF signal shift and the OFF-to-ON signal shift, while the combination as the first detection pattern and the combination as the second detection pattern are different from each other.

21

Alternatively, each of the first and second detection patterns may be a combination of ones selected from a group consisting of the ON-to-OFF signal shift, the OFF-to-ON signal shift, the turn-on time and the turn-off time, while the combination as the first detection pattern and the combination as the second detection pattern are different from each other. In this regard, referring to FIG. 23, the only requirement is that a combination of an ON-to-OFF shifts count B1, an OFF-to-ON shifts count B2, a time B3 and a time B4 during movement of the piston rod 37 in the first direction D1 is different from a combination of an ON-to-OFF shifts count F1, an OFF-to-ON shifts count F2, a time F3 and a time F4 during movement of the piston rod 37 in the second direction D2.

An example of the combinations is as follows:

First direction D1: B1 is 2; B2 is 2; B3 is 2 seconds, and B4 is 2 seconds.

Second direction D2: F1 is 1; F2 is 1; F3 is 2 seconds, and F4 is 4 seconds.

Another embodiment in which the bucket 24 is determined whether it is on the dump side E3 or the crowd side E4 and then the swing position of the bucket 24 is calculated will now be described.

As mentioned above, the bucket cylinder C5 is hydraulically controlled by the bucket control valve 72 which is operated by an electric signal outputted from the controller 71. Therefore, it can be grasped by means of the controller 71 whether the bucket 24 is moved to the dump side E3 or the crowd side E4.

It is assumed that the voltage of the potentiometer serving as the angle sensor 81 becomes low when the bucket 24 is in the vicinity of the end of its swing range end, and the voltage becomes high when the bucket 24 is in the vicinity of the neutral position 80. On this assumption, if the voltage of the angle sensor 81 varies in the increasing direction during the swing of the bucket 24 in the dump direction Y1 by means of the first solenoid 72d excited by the controller 71, the bucket 24 is determined as being on the crowd side E4 with respect to the neutral position 80. If the voltage of the angle sensor 81 varies in the reducing direction during the swing of the bucket 24 in the dump direction Y1, the bucket 24 is determined as being on the dump side E3 with respect to the neutral position 80. If the voltage of the angle sensor 81 varies in the increasing direction during the swing of the bucket 24 in the crowd direction Y2 by means of the second solenoid 72e excited by the controller 71, the bucket 24 is determined as being on the dump side E3 with respect to the neutral position 80. If the voltage of the angle sensor 81 varies in the reducing direction during the swing of the bucket 24 in the crowd direction Y2, the bucket 24 is determined as being on the crowd side E4 with respect to the neutral position 80.

As mentioned above, based on the voltage of the angle sensor 81 and the electric signal for operating the bucket control valve 72, it can be determined whether the bucket 24 is on the dump side E3 or the crowd side E4.

As a modification, the operation member 75 for operating the bucket 24 may be used. The controller 71 obtains an electric signal from the operation member 75. Therefore, when the lever 76 is operated in one direction to swing the bucket 24 in the dump direction Y1, if the voltage of the angle sensor 81 varies in the increasing direction, the bucket 24 is determined as being on the crowd side E4 with respect to the neutral position 80, and if the voltage of the angle sensor 81 varies in the reducing direction, the bucket 24 is determined as being on the dump side E3. When the lever 76 is operated in another direction to swing the bucket 24 in the

22

crowd direction Y2, if the voltage of the angle sensor 81 varies in the increasing direction, the bucket 24 is determined as being on the dump side E3 with respect to the neutral position 80, and if the voltage of the angle sensor 81 varies in the reducing direction, the bucket 24 is determined as being on the crowd side E4 with respect to the neutral position 80.

In this way, based on the voltage of the angle sensor 81 and the electric signal which the controller 71 obtains from the operation member 75, it can be determined whether the bucket 24 is on the dump side E3 or the crowd side E4.

These manners can be used without the detection device 41, thereby reducing costs.

The working machine 1 according to the embodiments has the following effects:

The working machine 1 comprises the support member (or the arm 23), the supported member (or the working tool 24) pivotally supported by the first end portion of the support member (or the arm 23) via the pivot, and the cylinder (or the working tool cylinder C5) of which telescopic movement causes the supported member (or the working tool 24) to swing around the pivot (or the working tool shaft 29). The cylinder (or the working tool cylinder C5) includes the cylinder tube 36, and the piston rod 37 inserted into the cylinder tube 36. The cylinder (or the working tool cylinder C5) is constituted by the hydraulic cylinder telescopically moved by extension and contraction movement of the piston rod 37 with respect to the cylinder tube 36 by hydraulic fluid flowing through the fluid passage formed in the piston rod 37. The cylinder tube 36 is pivotally supported by one of the first and second end portions of the support member (or the arm 23), and the piston rod 37 is pivotally supported by the other of the first and second end portions of the support member (or the arm 23). The detection device 41 for detecting the telescopic movement state of the cylinder (or the working tool cylinder C5) is provided between the cylinder (or the working tool cylinder C5) and the support member (or the arm 23).

Due to the above configuration, the cylinder (or the working tool cylinder C5) is constituted by the hydraulic cylinder telescopically moved by hydraulic fluid flowing through the fluid passage formed in the piston rod 37 so that the cylinder (or the working tool cylinder C5) and the support member (or the arm 23) have a space therebetween without a pipe or a hose. The detection device 41 disposed in the space is advantageously prevented from being damaged in comparison with the detection device 41 when provided on a portion pivotally supporting the supported member (or the working tool 24). The space is effectively usable so that the detection device 41 disposed in the space is prevented or reduced from spoiling the appearance of the working machine, and from increasing the size thereof.

The working machine 1 may comprise the machine body 2 and the boom 22 swingably attached to the machine body 2. The support member may be the arm 23 pivotally supported at the basal end portion thereof by the boom 22. The supported member may be the working tool 24 pivotally supported by the tip end portion of the arm 22 via the pivot (or the working tool shaft 29). The cylinder may be the working tool cylinder C5 for swinging the working tool 24 around the pivot.

Due to the above configuration in which the working tool cylinder C5 is constituted by the hydraulic cylinder telescopically moved by hydraulic fluid flowing through the fluid passage in the piston rod 37, the working tool cylinder C5 and the arm 23 have a space therebetween without a pipe

23

or a hose. Therefore, the detection device 41 disposed in the space is prevented or reduced from being damaged.

The detection device 41 includes the detected object 42 provided on one of the cylinder tube 36 and the piston rod 37, and the detector 43 provided on the other of the cylinder tube 36 and the piston rod 37. The detector 43 detects the detected object 42 for detection of the telescopic movement state of the cylinder.

Due to the above configuration, the detection device 41 can be compactly disposed between the working tool cylinder C5 and the arm 23.

The detected object 42 is attached onto the outer surface of the cylinder tube 36 facing the arm 23. The detected object 42 has the predetermined length in the longitudinal direction of the cylinder tube 36, and includes the magnets 47A and 47B. The detector 43 is constituted by the proximity sensor, which is attached to the piston rod 37 between the cylinder tube 36 and the arm 23 so as to be able to face the detected object 42, and which is configured to move together with the piston rod 37 to detect the magnet 47A or 47B.

Due to the above configuration, the detection device 41 can be configured simply and economically.

The working machine 1 comprises the detector mount 52 to which the detector 43 is attached, and the connection mechanism 53 connecting the detector mount 52 to the piston rod 37. The piston rod 37 includes the first connection piece 59 joined to the piston rod 37, the second connection piece 60 joined to the detection attachment member 52, and the connection pin 61 passed through the first connection piece 59 and the second connection piece 60 so as to connect the first connection piece 59 and the second connection piece 60 to each other. The pin hole (or the first pin hole 64) is formed in the connection piece 59 to allow the connection pin 61 to pass therethrough. The pin hole is formed into the long hole shape to allow axial rotation of the piston rod 37.

Due to the above configuration, the simple connection mechanism 53 prevents the piston rod 37 from being damaged because of the axial rotation of the piston rod 37.

The working machine 1 comprises the arm 23, the bucket (or the working tool) 24 pivotally and swingably supported by the tip end portion of the arm 23, the bucket cylinder (or the working tool cylinder) C5 pivotally supported by the arm 23 via the cylinder shaft 35, the telescopic movement of the bucket cylinder C5 causing the bucket 24 to swing, the angle sensor 81 which detects the swing angle G1 of the bucket cylinder C5 when the bucket 24 is disposed on the dump side E3 with respect to the neutral position 80 where the swing angle G of the bucket cylinder C5 around the cylinder shaft 35 becomes maximum, and which detects the swing angle G2 of the bucket cylinder C5 when the bucket 24 is disposed on the crowd side E4 with respect to the neutral position 80, the detection device 41 which detects ON and OFF signals indicating whether the bucket cylinder C5 is extended further from the neutral position 80 or is contracted further from the neutral position 80, and the controller 71 for determining the swing position of the bucket 24 based on the swing angle G detected by means of the angle sensor 81 and the result of detection by means of the detection device 41. The controller 71 determines whether the bucket 24 is on the dump side E3 or the crowd side E4 in accordance with the first detection pattern appearing in detection of the ON and OFF signals by means of the detection device 41 when the bucket 24 is moved in the first direction D1 from the dump side E3 to the crowd side E4, and the second detection pattern appearing in detection of the ON and OFF signals by

24

means of the detection device 41 when the bucket 24 is moved in the second direction D2 from the crowd side E4 to the dump side E3.

Due to the above configuration, the swing position of the working tool 24 can be accurately calculated based on the swing angle of the bucket cylinder C5 detected by means of the angle sensor 81 and the result of detection by means of the detection device 41.

Based on the first detection pattern and the second detection pattern, the controller 71 determines whether the bucket 24, when in the predetermined range as the vicinity of the neutral position 80, is on the dump side E3 or the crowd side E4.

Due to the above configuration, the position of the bucket 24 in the predetermined range as the vicinity of the neutral position 80 where the variation direction of the swing angle G is reversed can be determined while the angle sensor 81 alone is insufficient to determine the position.

While the bucket 24 is disposed out of the predetermined range, the controller 71 keeps the result of the determination regarding the position of the bucket 24 when in the predetermined range.

Due to this configuration, the detection device 41 can be configured to detect the bucket 24 in only the predetermined range as the vicinity of the neutral position 80.

The bucket cylinder 24 includes the cylinder tube 36, and the piston rod 37 extended and contracted with respect to the cylinder tube 36. The detection device 41 includes the detected object 42 provided on one of the cylinder tube 36 and the piston rod 37, and the detector 43 provided on the other of the cylinder tube 36 and the piston rod 37. The detector 43 outputs the on or off signal in response to whether the detected object 42 is detected or not.

Due to the above configuration, the position of the piston rod 37 with respect to the cylinder tube 36 facilitates the determination of the position of the bucket 24.

Either the range ranging from the neutral position 80 to the intermediate position between the neutral position 80 and the end position on the dump side E3 or the range ranging from the neutral position 80 to the intermediate position between the neutral position 80 and the end position on the crowd side E4 is defined as the detectable range 87 where the detected object 42 is detectable by means of the detector 43.

Due to the above configuration, the length of the detected object 42 can be reduced, thereby minimizing the detection device 41.

Each of the first detection pattern and the second detection pattern is a combination of ON-to-OFF and OFF-to-ON signal shifts. The combination as the first detection pattern and the combination as the second detection pattern are different from each other.

Due to the above configuration, the detection device has many variations.

When operation of the bucket 24 is ended, the controller 71 stores the result of determination of whether the bucket 24 is on the dump side E3 or the crowd side E4. The controller 71 defines the position of the bucket 24 stored on the ending of operation of the bucket 24 as the initial position for restarting of operation of the working tool, and performs the determination when the bucket cylinder C5 is extended or contracted from the initial position.

Due to the above configuration, the position of the bucket 24 does not have to be confirmed every restarting of operation of the bucket 24 after ending of the last operation of the bucket 24, thereby enhancing the convenience in operation of the bucket 24.

25

The working machine comprises the display unit **90** connected to the controller **71**. The display unit **90** displays the message for urging an operator to operate to locate the bucket **24** on either the dump side **E3** or the crowd side **E4**.

Due to the above configuration, the operator can be informed of a position change of the working tool **24**, for example.

The controller **71** performs the determination based on the detection value outputted from the angle sensor **81** when the ON/OFF signal shift occurs in detection by means of the detection device **41**.

Due to this configuration, whether the bucket **24** is on the dump side **E3** or the crowd side **E4** can be determined based on the detection value from the angle sensor **81** when the ON/OFF signal shift occurs in detection by means of the detection device **41**.

When operation of the bucket **24** is ended, the controller **71** stores information indicating whether the bucket **24** is on the dump side **E3** or the crowd side **E4**. When operation of the bucket **24** is restarted, the controller **71** determines the swing position of the bucket **24** based on the information stored on the last ending of operation of the bucket **24**.

Due to this configuration, the position of the bucket **24** does not have to be confirmed every restarting of operation of the bucket **24** after ending of the last operation of the bucket **24**, thereby enhancing the convenience in operation of the bucket **24**.

The working machine comprises the instruction input unit (or the switch **85**) for receiving the instruction input from an operator, the instruction input indicating whether the bucket **24** is on the dump side **E3** or the crowd side **E4**.

Due to this configuration, the operator can inform the controller **71** regarding whether the bucket **24** is on the dump side **E3** or the crowd side **E4**.

While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A working machine comprising:

a machine body;

a boom swingably attached to the machine body;

a support member pivotally supported at a basal end portion thereof by the boom;

a supported member supported by a tip end portion of the support member via a pivot, the tip end portion being an opposite side of the basal end portion; and

a cylinder of which telescopic movement causes the supported member to swing around the pivot, the cylinder including

a cylinder tube, and

a piston rod inserted into the cylinder tube, and

a linkage including a first link and a second link, the first link being pivotally supported at a first end thereof by the tip end portion of the support member, the second link being pivotally supported at a first end thereof by the supported member and pivotally connected at a second end thereof to a second end of the first link, wherein the cylinder

includes a rod head provided at a tip end portion of the piston rod and having ports via which hydraulic fluid is supplied or discharged,

is constituted by a hydraulic cylinder telescopically moved by extension and contraction movement of the piston rod with respect to the cylinder tube by

26

hydraulic fluid supplied to or discharged from fluid passages formed in the piston rod via the ports, is positioned such that a length of the cylinder extends along the support member and the cylinder is opposed to the support member, and

includes an attachment portion provided at a bottom side end portion of the cylinder tube, the bottom side end portion being a side opposite to a side at which the extension movement of the piston rod is performed, the attachment portion being pivotally supported by the second end of the first link and the second end of the second link, and

has the rod head at the tip end portion of the piston rod pivotally supported by the basal end portion of the support member, and

at a position closer to the basal end portion of the support member than the first link, a detection device for detecting a telescopic movement state of the cylinder is provided to intervene between a surface of the cylinder facing the support member and a surface of the support member facing the cylinder,

the detection device includes a detected object and a detector, the detector detecting the detected object, thereby detecting the telescopic movement state,

one of the detected object and the detector is attached to an outer surface of the cylinder tube facing the support member, and

the other of the detected object and the detector is disposed between the cylinder tube and the support member so as to be able to face the one of the detected object and the detector, and is attached to an attachment structure joined to the rod head.

2. The working machine according to claim 1, wherein the support member is an arm pivotally supported at the basal end portion thereof by the boom, wherein the supported member is a working tool pivotally supported by a tip end portion of the arm via the pivot, and

wherein the cylinder is a working tool cylinder for swinging the working tool around the pivot.

3. The working machine according to claim 1,

wherein the detection device includes

the detected object provided on the cylinder tube, and the detector attached to the attachment structure.

4. The working machine according to claim 3,

wherein the detected object is attached onto the outer surface of the cylinder tube facing the support member, wherein the detected object has a predetermined length in a longitudinal direction of the cylinder tube, and includes a magnet, and

wherein the detector is constituted by a proximity sensor, which is attached to the piston rod between the cylinder tube and the support member so as to be able to face the detected object, and which is configured to move together with the piston rod to detect the magnet.

5. The working machine according to claim 3,

wherein the attachment structure includes

a detector mount to which the detector is attached; and a connection mechanism connecting

the detector mount to the piston rod,

wherein the connection mechanism includes

a first connection piece joined to the rod head,

a second connection piece joined to the detector mount,

and a connection pin passed through the first connection piece and the second connection piece so as to connect the first connection piece and the second connection piece to each other, and

27

wherein a pin hole is formed in the first connection piece to allow the connection pin to pass through, the pin hole allowing axial rotation of the piston rod.

6. The working machine according to claim 1, further comprising:

an arm serving as the support member;
a working tool serving as the supported member pivotally supported by a tip end portion of the arm;
a working tool cylinder serving as the cylinder pivotally supported by the arm via a cylinder shaft, a telescopic movement of the working tool cylinder causing the working tool to swing;

an angle sensor which detects a swing angle of the working tool cylinder when the working tool is disposed on a dump side with respect to a neutral position where a swing angle of the working tool cylinder around the cylinder shaft becomes maximum, and which detects a swing angle of the working tool cylinder when the working tool is disposed on a crowd side with respect to the neutral position; and

a controller for determining a swing position of the working tool based on the swing angle detected by means of the angle sensor and a result of detection by means of the detection device,

wherein the detection device detects ON and OFF signals indicating whether the working tool cylinder is extended further from the neutral position or is contracted further from the neutral position, and

wherein the controller determines whether the working tool is on the dump side or the crowd side in accordance with a first detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a first direction from the dump side to the crowd side, and a second detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a second direction from the crowd side to the dump side.

7. A working machine comprising:

a machine body;
a boom swingably attached to the machine body,
an arm pivotally supported at a basal end portion thereof by the boom;

a working tool supported by a tip end portion of the arm via a pivot, the tip end portion being an opposite side of the basal end portion;

a linkage including a first link and a second link, the first link being pivotally supported at a first end thereof by the tip end portion of the arm, the second link being pivotally supported at a first end thereof by the working tool and pivotally connected at a second end thereof to a second end of the first link;

a working tool cylinder including a cylinder tube and a piston rod extended and contracted with respect to the cylinder tube, the working tool cylinder

having a first end in a longitudinal direction pivotally supported by the second end of the first link and the second end of the second link, and having a second end in the longitudinal direction pivotally supported by the basal end portion of the arm via a cylinder shaft,

being configured to have a swing angle that, when the working tool cylinder is being extended from a fully contracted state to a fully extended state, gradually increases and becomes maximum in a halfway point, and then gradually reduces and the working tool

28

cylinder reaches the fully extended state, the swing angle being defined as an angle of the working tool cylinder swinging around the cylinder shaft,

wherein a telescopic movement of the working tool cylinder in the longitudinal direction causing the working tool to swing in a crowd direction, which is a direction approaching the arm, and to swing around the pivot in a dump direction, which is a direction opposite to the crowd direction;

an angle sensor which detects the swing angle of the working tool cylinder around the cylinder shaft, the angle sensor detecting the swing angle of the working tool cylinder when the working tool is disposed on a dump side with respect to a neutral position, and detecting the swing angle of the working tool cylinder when the working tool is disposed on a crowd side with respect to the neutral position, the neutral position being a swing position of the working tool in a state where the swing angle of the working tool cylinder is at the maximum;

a detection device including:

a detected object provided on one of the cylinder tube and the piston rod; and

a detector provided on the other of the cylinder tube and the piston rod wherein the detector detects the detected object,

the detection device detecting, in response to whether the detected object is detected by the detector or not, ON and OFF signals indicating whether the working tool cylinder is extended past the state where the swing angle is at the maximum or is contracted further from the state where the swing angle is at the maximum; and

a controller for determining a swing position of the working tool based on the swing angle detected by means of the angle sensor and a result of detection by means of the detection device,

wherein the controller determines

whether the working tool is on the dump side or the crowd side in accordance with a first detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a first direction from the dump side to the crowd side, and a second detection pattern appearing in detection of the ON and OFF signals by means of the detection device when the working tool is moved in a second direction from the crowd side to the dump side, and

based on a result of the determination whether the working tool is on the dump side or the crowd side and the swing angle of the working tool cylinder detected by the angle sensor, a location of the working tool in a swing range, which is a range where the working tool swings around the pivot.

8. The working machine according to claim 7,

wherein, based on the first detection pattern and the second detection pattern, the controller determines, when the working tool is within a predetermined range in the vicinity of the neutral position, whether the working tool is on the dump side or the crowd side with respect to the neutral position.

9. The working machine according to claim 8,

wherein when the working tool is out of the predetermined range, the controller keeps a result of the determination performed within the predetermined range.

29

10. The working machine according to claim 7,
 wherein either a range ranging from the neutral position
 to an intermediate position between the neutral position
 and an end position on the dump side or a range ranging
 from the neutral position to an intermediate position
 between the neutral position and an end position on the
 crowd side is defined as a detectable range where the
 detected object is detectable by means of the detector. 5
11. The working machine according to claim 7,
 wherein each of the first detection pattern and the second
 detection pattern is a combination of ON-to-OFF and
 OFF-to-ON signal shifts, and 10
- wherein the combination as the first detection pattern and
 the combination as the second detection pattern are
 different from each other. 15
12. The working machine according to claim 7,
 wherein when operation of the working tool is ended, the
 controller stores the result of determination of whether
 the working tool is on the dump side or the crowd side,
 and 20
- wherein the controller defines the position of the working
 tool stored on the ending of operation of the working
 tool as an initial position for restarting of operation of
 the working tool, and performs the determination when
 the working tool cylinder is extended or contracted 25
- from the initial position.

30

13. The working machine according to claim 7, further
 comprising:
 a display unit connected to the controller,
 wherein the display unit displays a message for urging an
 operator to operate to locate the working tool on either
 the dump side or the crowd side.
14. The working machine according to claim 7,
 wherein the controller performs the determination based
 on a detection value outputted from the angle sensor
 when an ON/OFF signal shift occurs in detection by
 means of the detection device.
15. The working machine according to claim 7,
 wherein when operation of the working tool is ended, the
 controller stores information indicating whether the
 working tool is on the dump side or the crowd side, and
 wherein when operation of the working tool is restarted,
 the controller determines the swing position of the
 working tool based on the information stored on the last
 ending of operation of the working tool.
16. The working machine according to claim 7, further
 comprising:
 an instruction input unit for receiving an instruction input
 from an operator, the instruction input indicating
 whether the working tool is on the dump side or the
 crowd side.

* * * * *